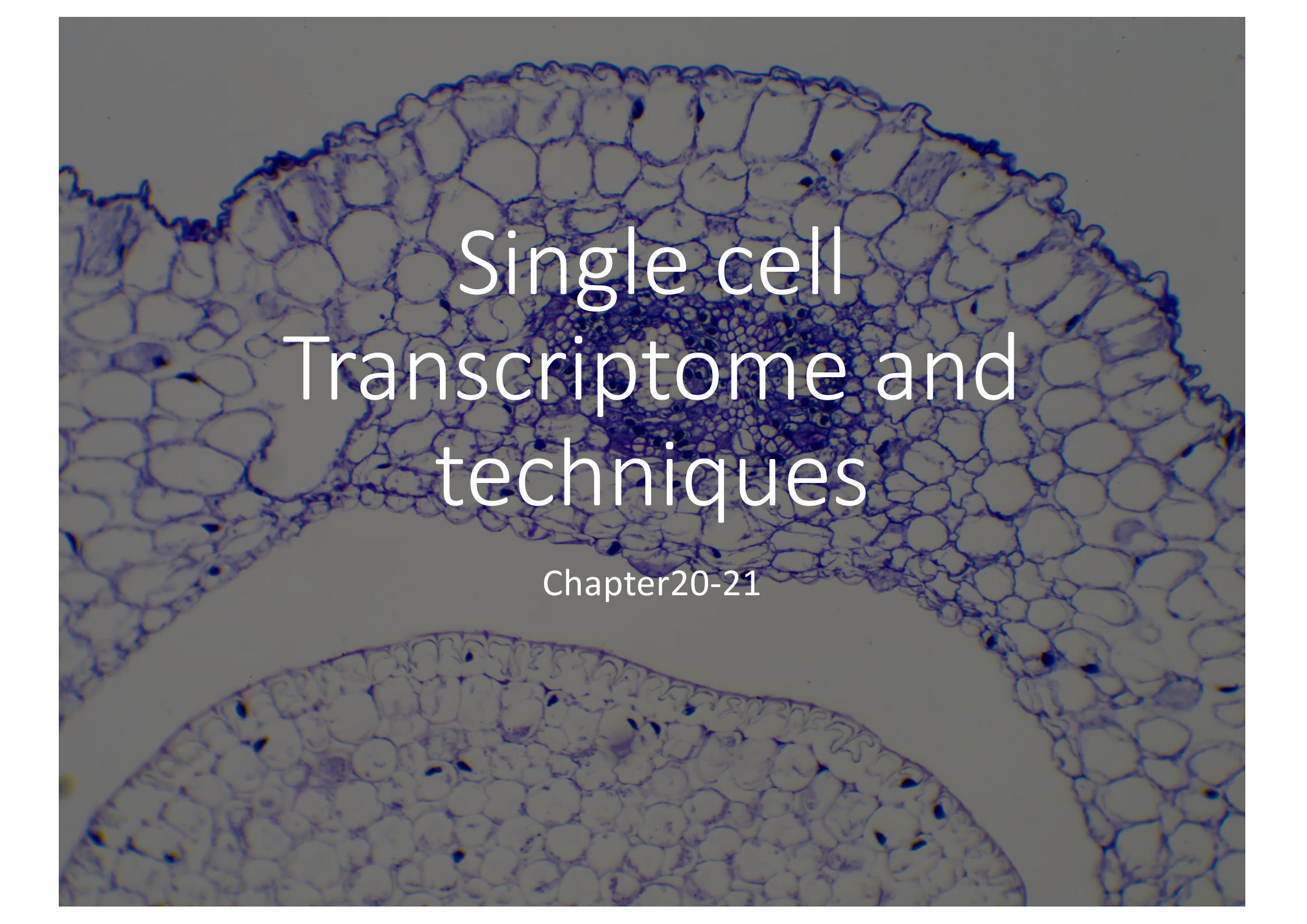


A microscopic image of a plant tissue cross-section, likely a leaf, showing a network of cells. The cells are stained with a blue dye, highlighting their cell walls and internal structures. The tissue is arranged in a somewhat circular or semi-circular pattern, with a central region that appears denser and more complex, possibly representing a vascular bundle or a specific tissue layer. The overall texture is granular and cellular.

Single cell Transcriptome and techniques

Chapter20-21

Content



Bulk RNA-seq



scRNA-seq



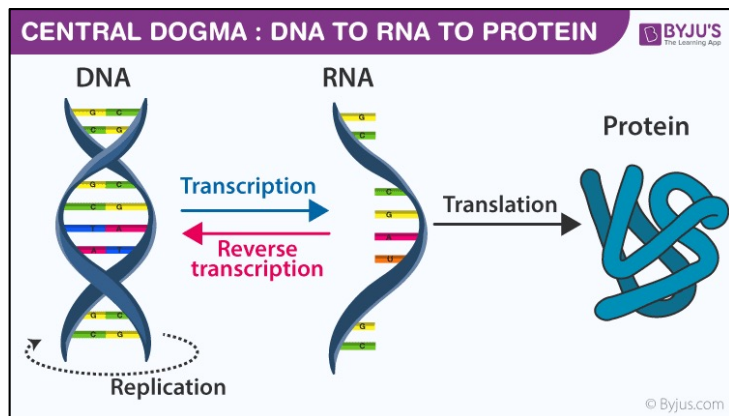
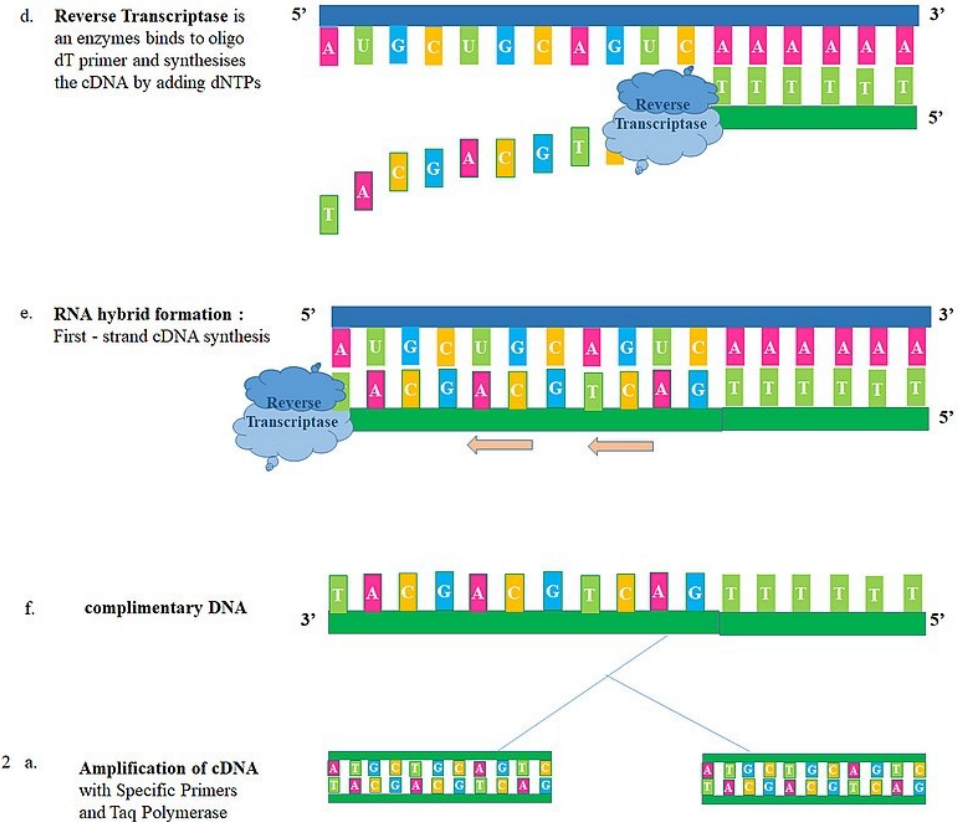
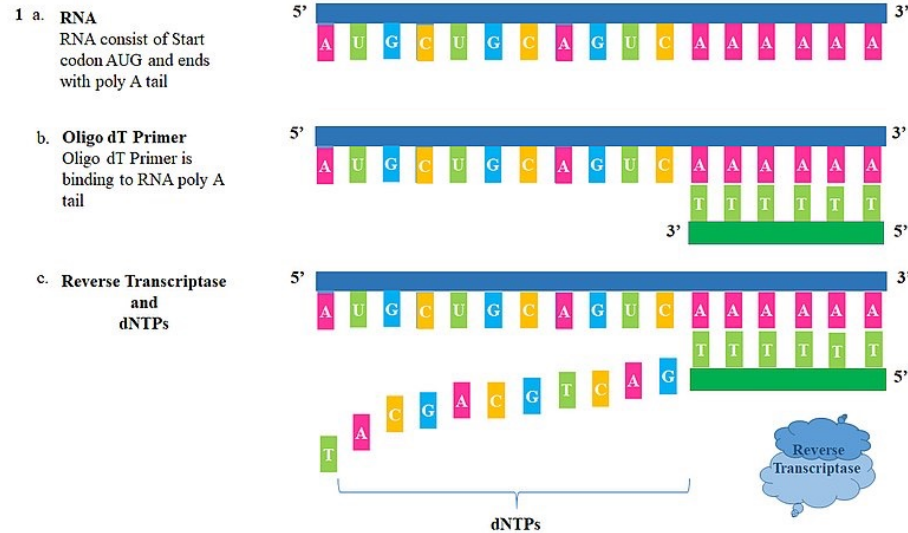
Single-molecule
RNA-seq



Spatial
transcriptome

RT-PCR

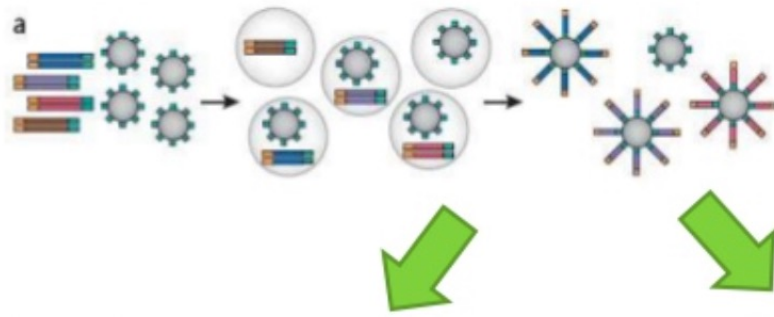
Single gene



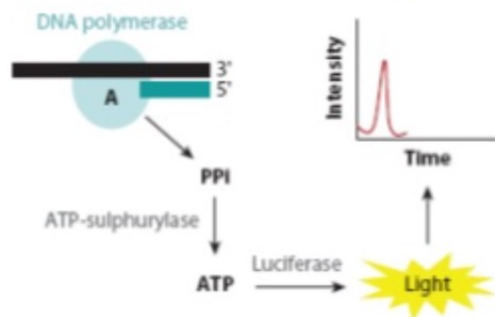
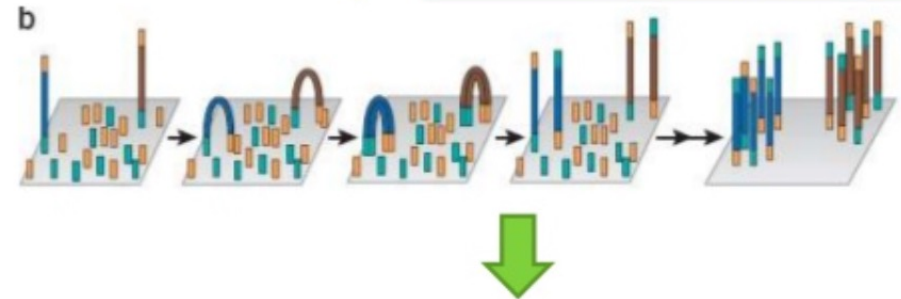
NGS WORKFLOW

Sample Extraction , DNA fragmentation and *invitro* adapter ligation

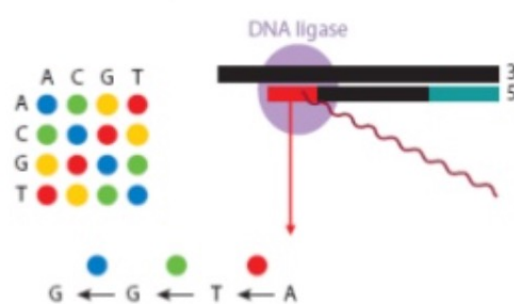
Clonal Amplification by
Emulsion PCR



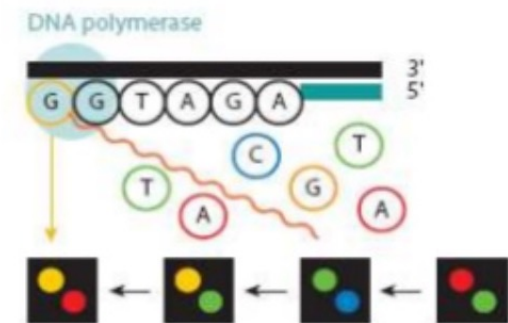
Clonal Amplification by
Bridge PCR



Pyrosequencing
(454 Sequencing)



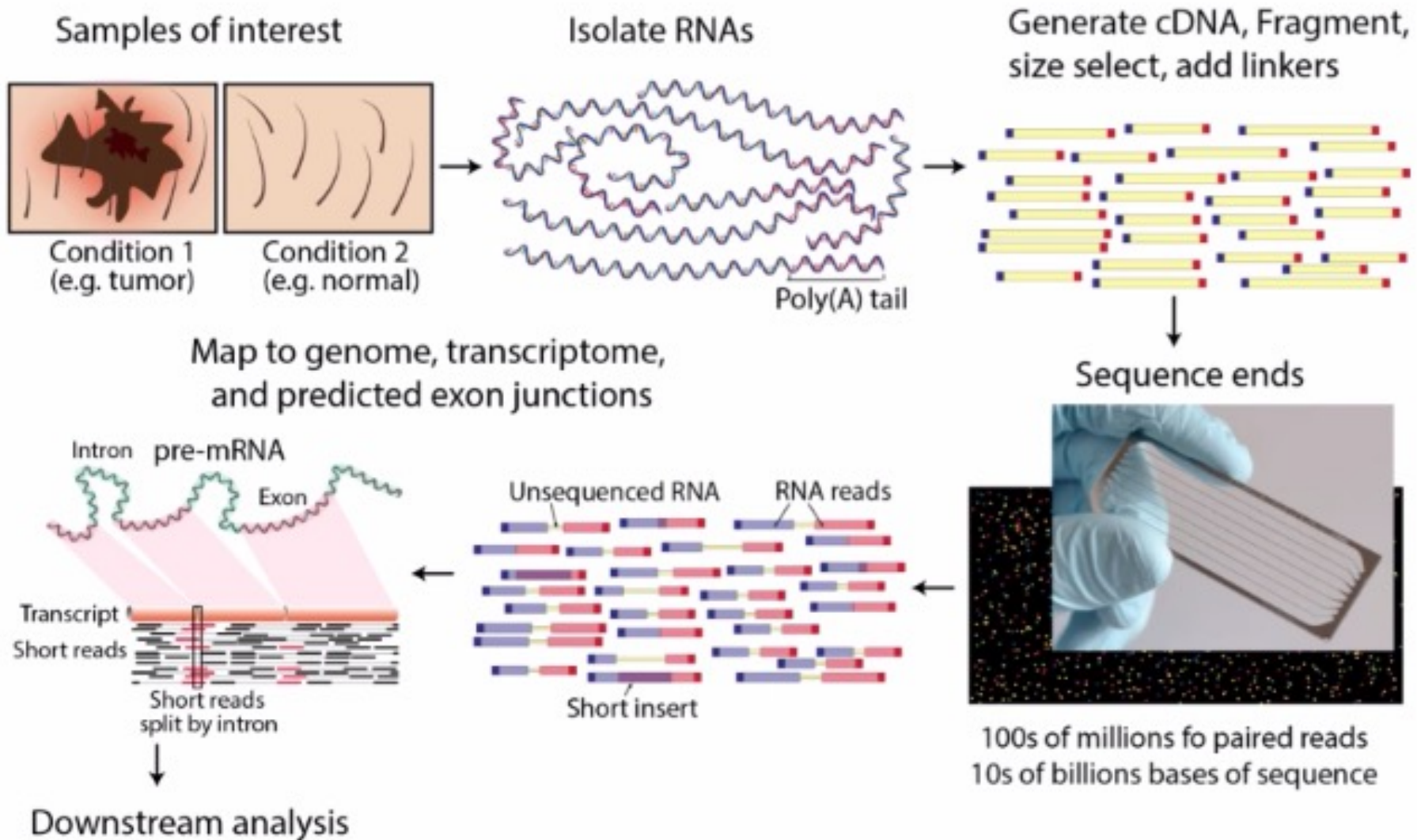
Sequencing-by-ligation
(SOLiD Platform)



Sequencing-by-synthesis
(Solexa Technology)

all genes

RNA sequencing



Understanding complexity

Whole Tissue/Organs
(Genetic) Disease Model



Complex Tissue



Bulk Genomics



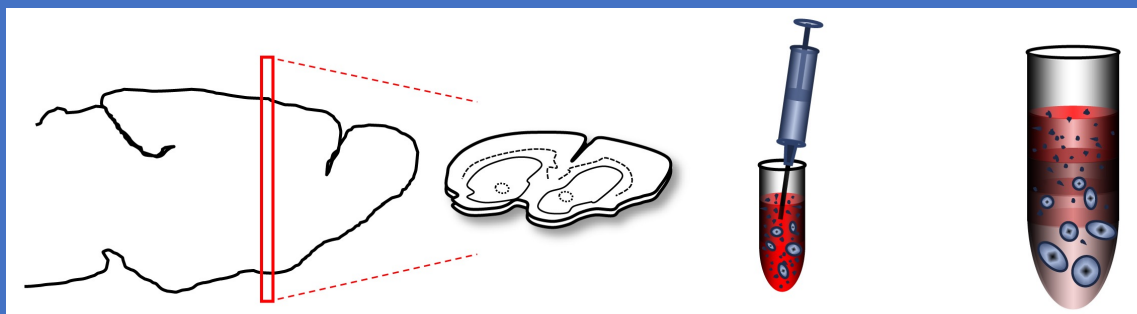
Flow
Cytometry +
Bulk
Genomics



Single Cell
Genomics
(+ Cytometry)

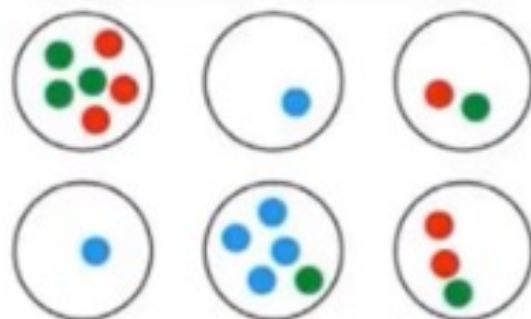


Spatial Transcriptomics



Why Single-Cell RNA-seq?

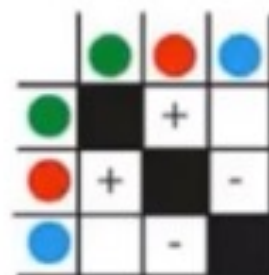
Single cell transcriptome analysis



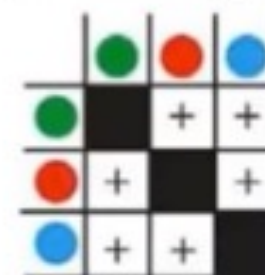
Bulk analysis

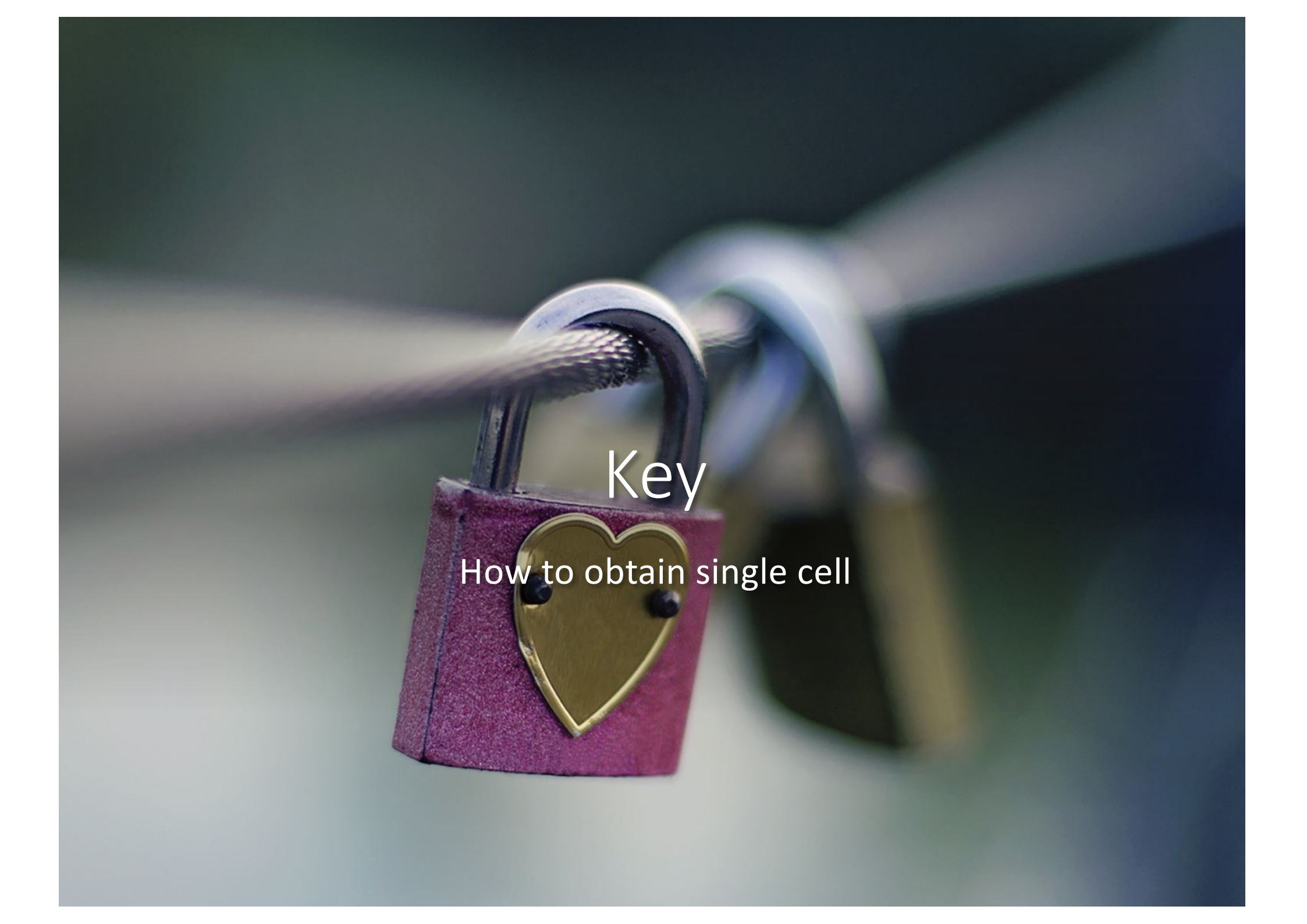


Coexpression Matrix
(single cell)



Coexpression Matrix
(bulk analysis)

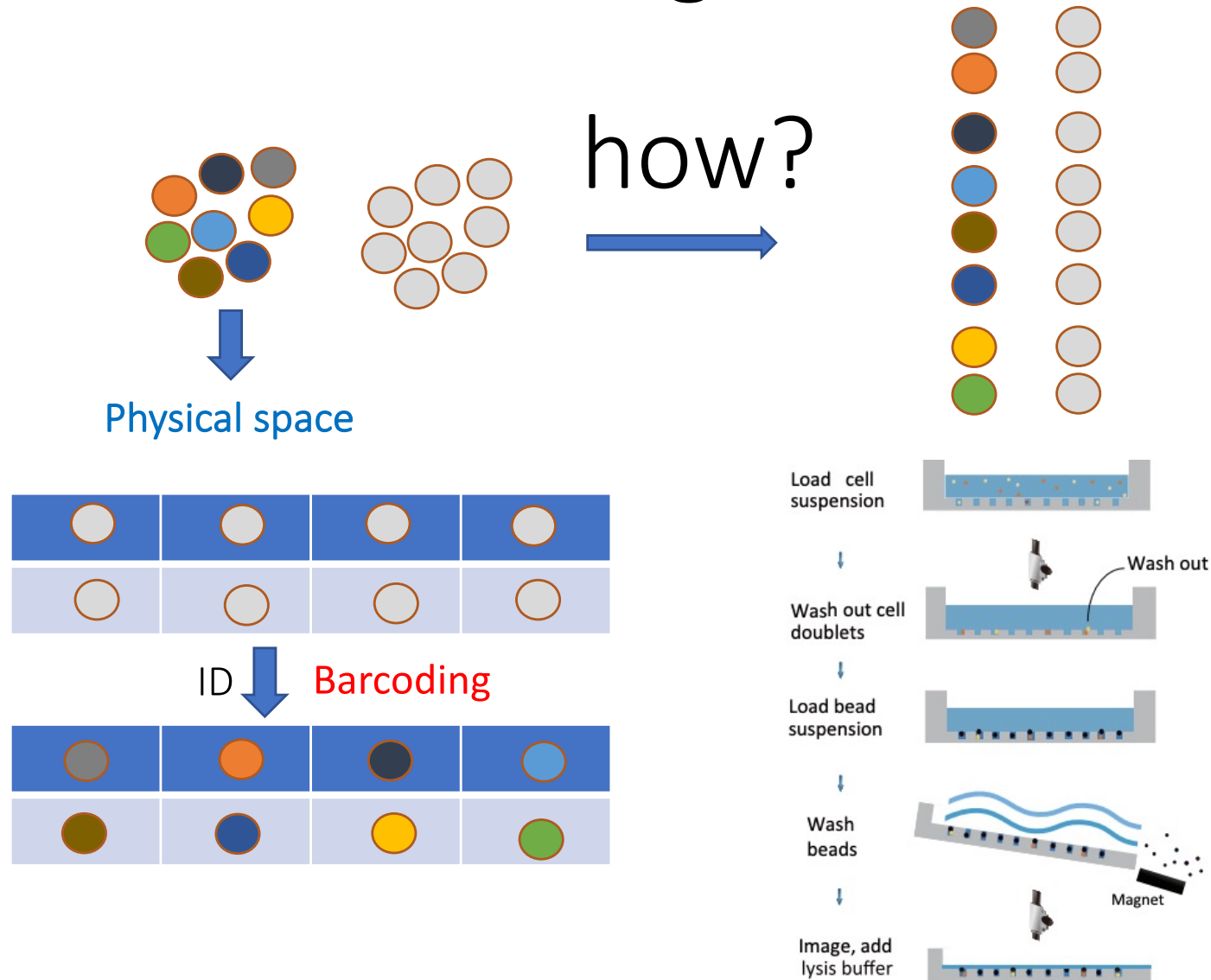




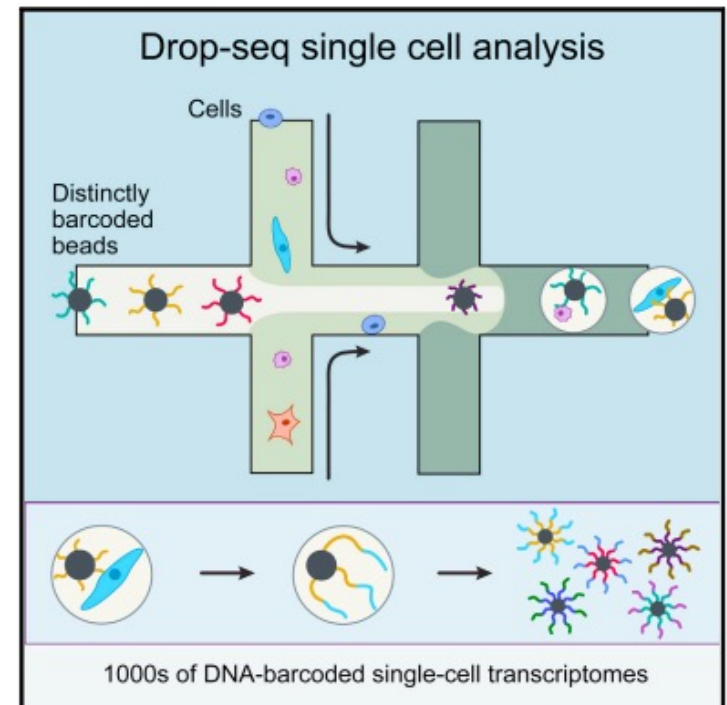
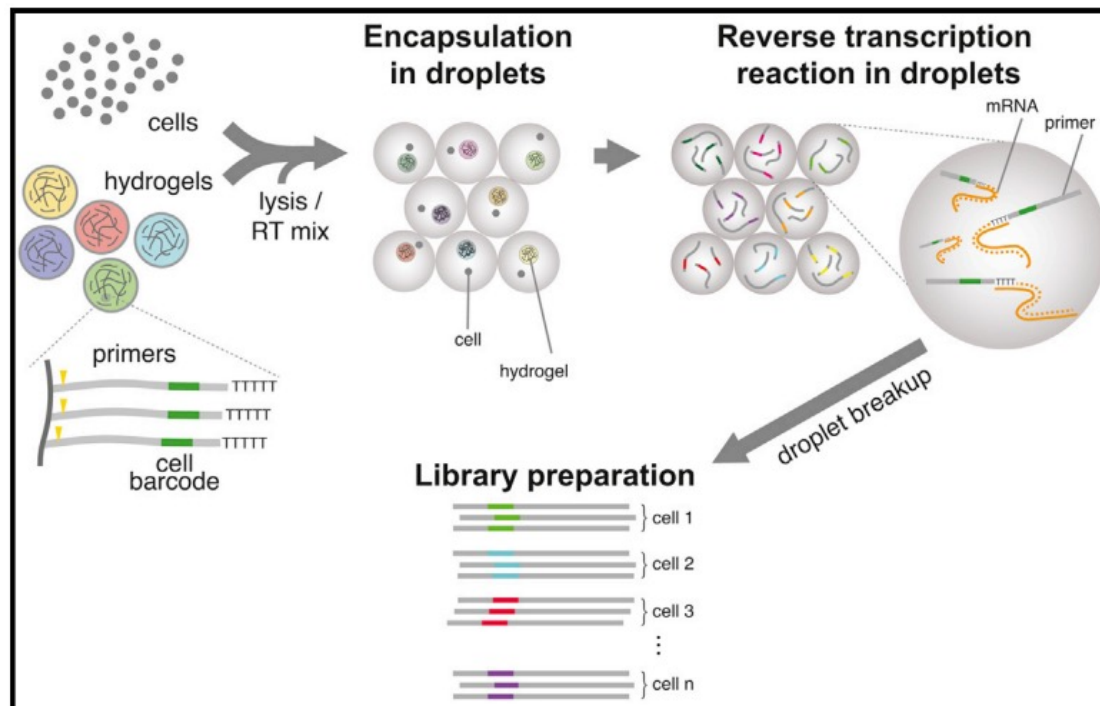
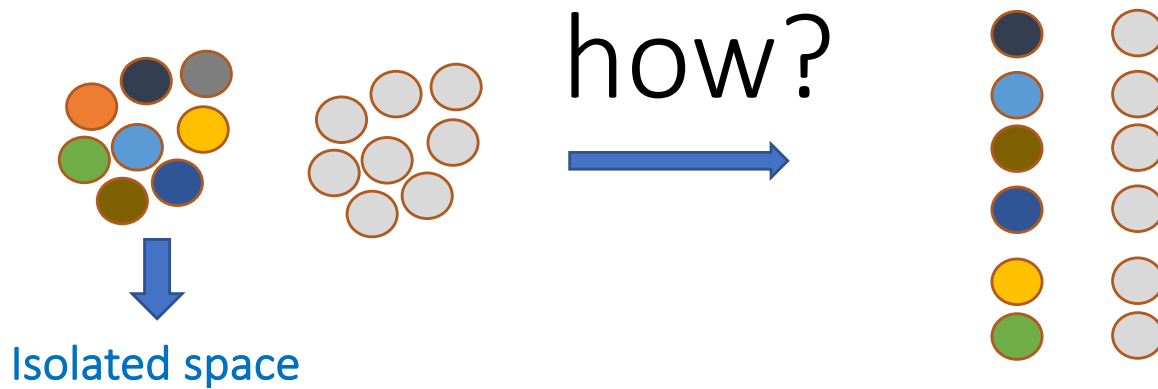
Key

How to obtain single cell

Bulk-cell to single-cell



Bulk-cell to single-cell



Barcoding

Key component

oligos for capture RNA

Barcode

oligos for compatible with sequencing flowcell

modification for release from beads

modification for immobilization

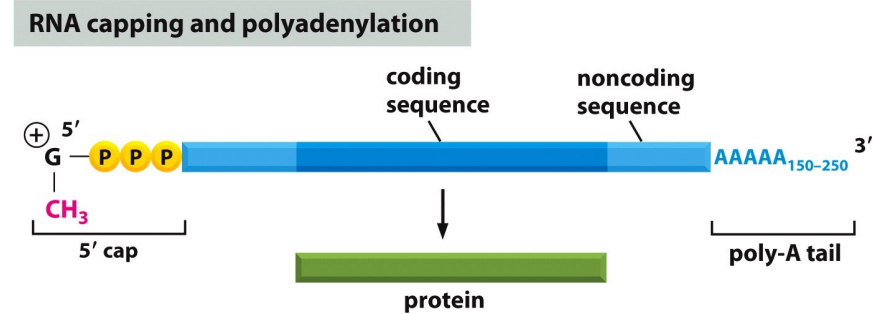
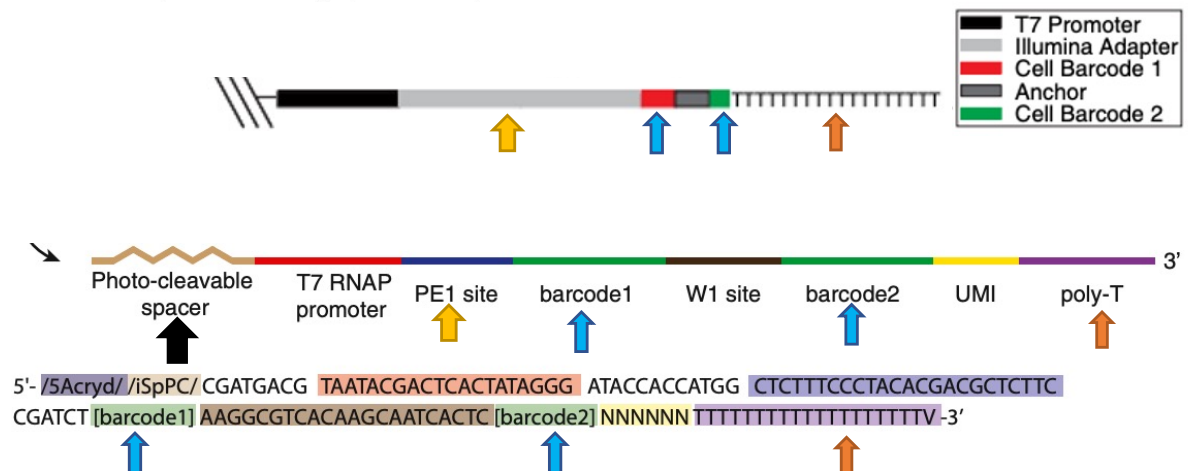
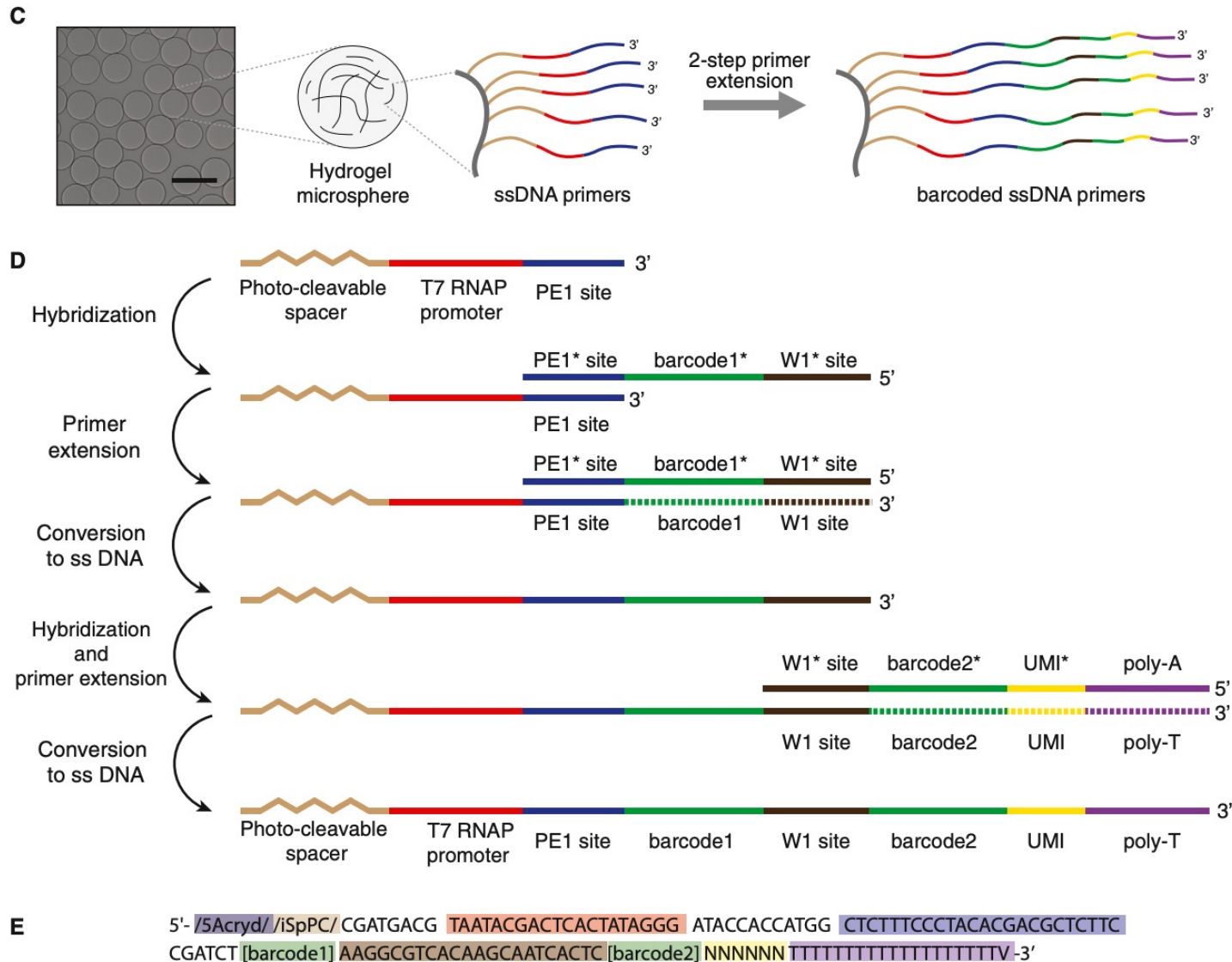


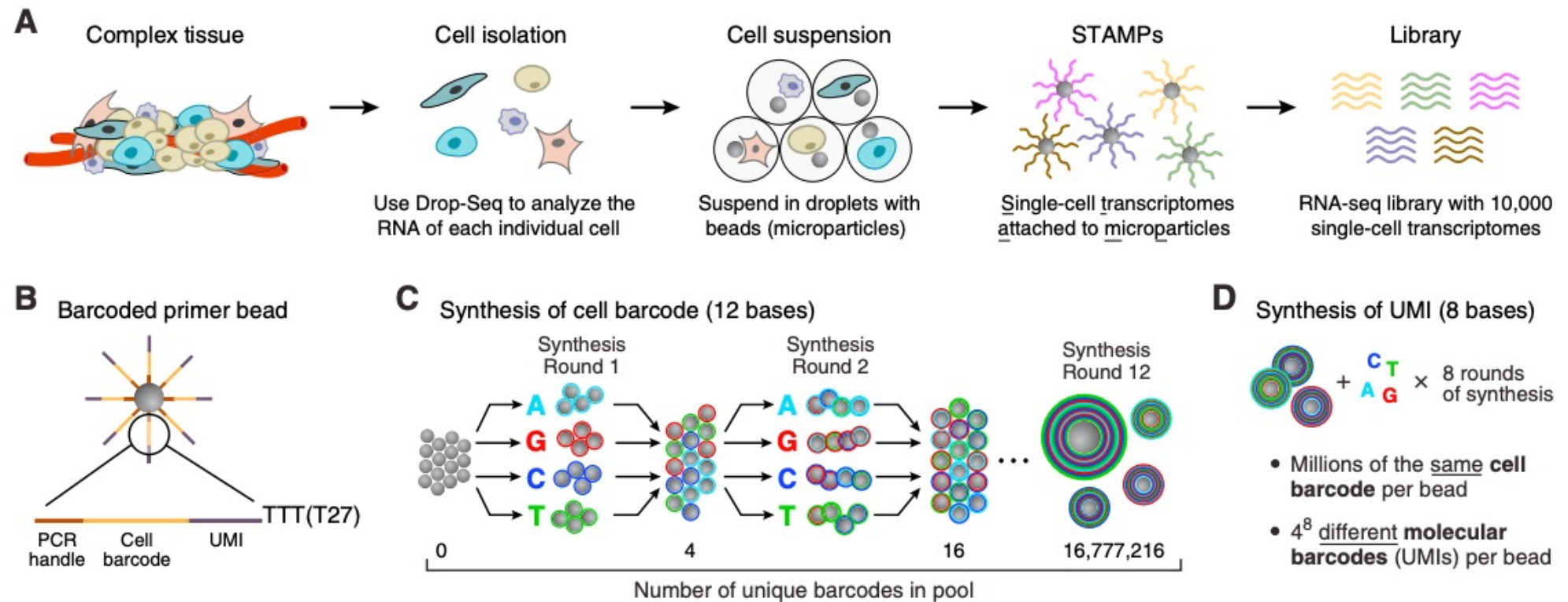
Figure 7-16a Essential Cell Biology 3/e (© Garland Science 2010)



Barcoding methods: Annealing



Barcoding methods: Synthesis

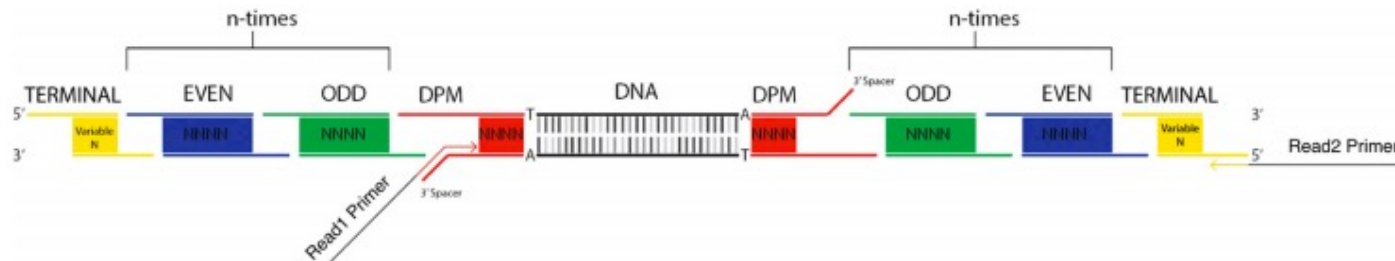


Barcoding methods: Ligation

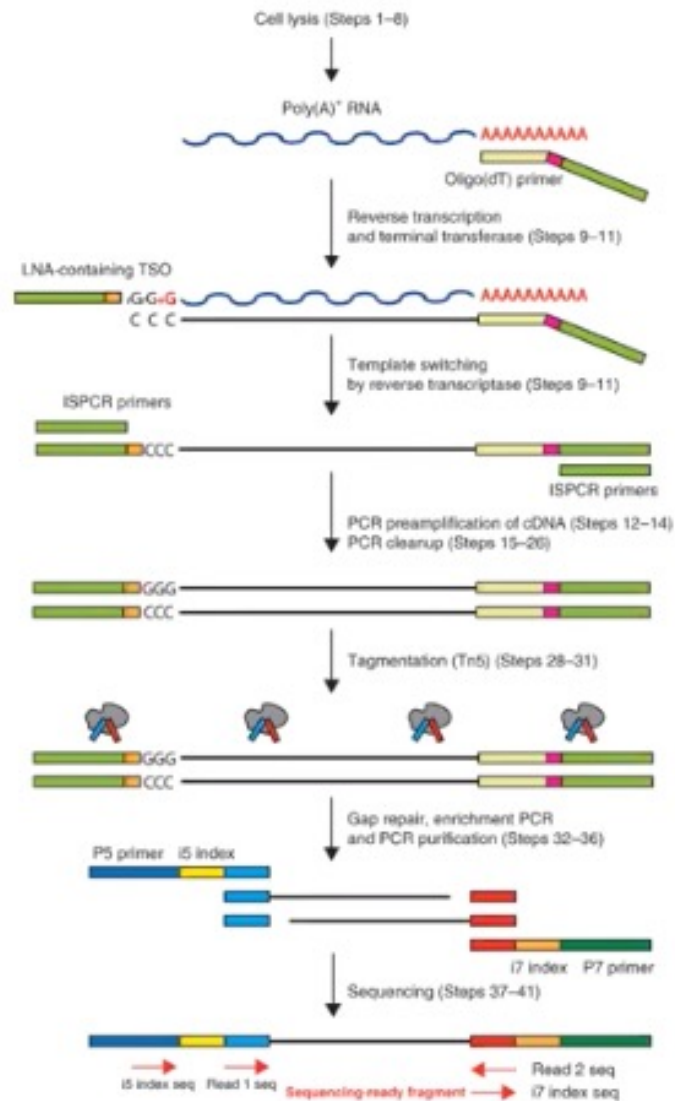
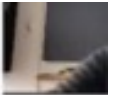


Final DNA structure

After SPRITE, the genomic DNA contains a DPM tag ligated on both ends as well as the Odd, Even, and Terminal tags. We call the full tag sequence a barcode. The product is represented below:



SMART-based

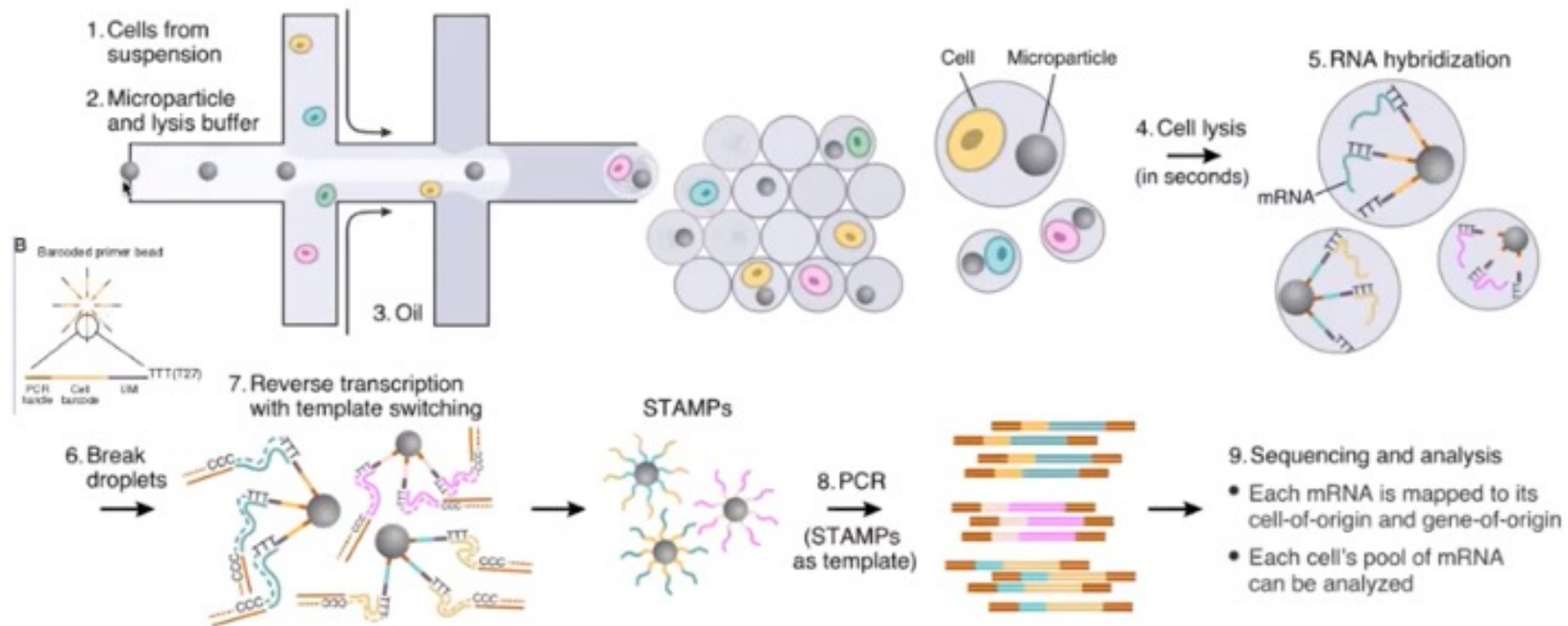


Features

- Developed for single cell but can be used for bulk tissues.
- Selects for poly-A tail.
- Full transcript assay.
 - Uses template switching for 5' end capture.
- Expensive to generate, so usually hundreds of cells.
- Better gene detection.
- Often do not use UMI.

Droplet-based

- Tens of thousands cells per experiment
- 10X Genomics commercial solution

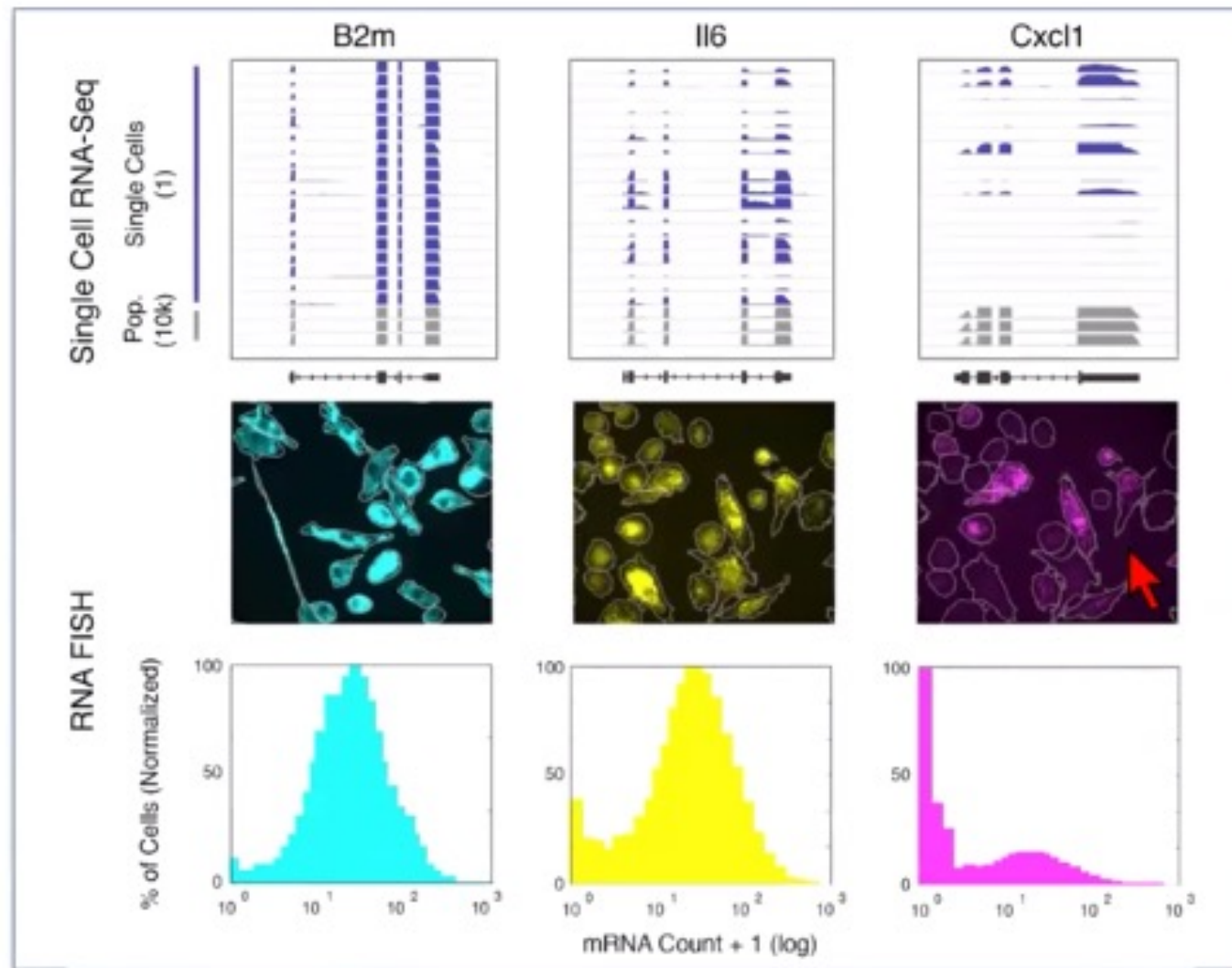




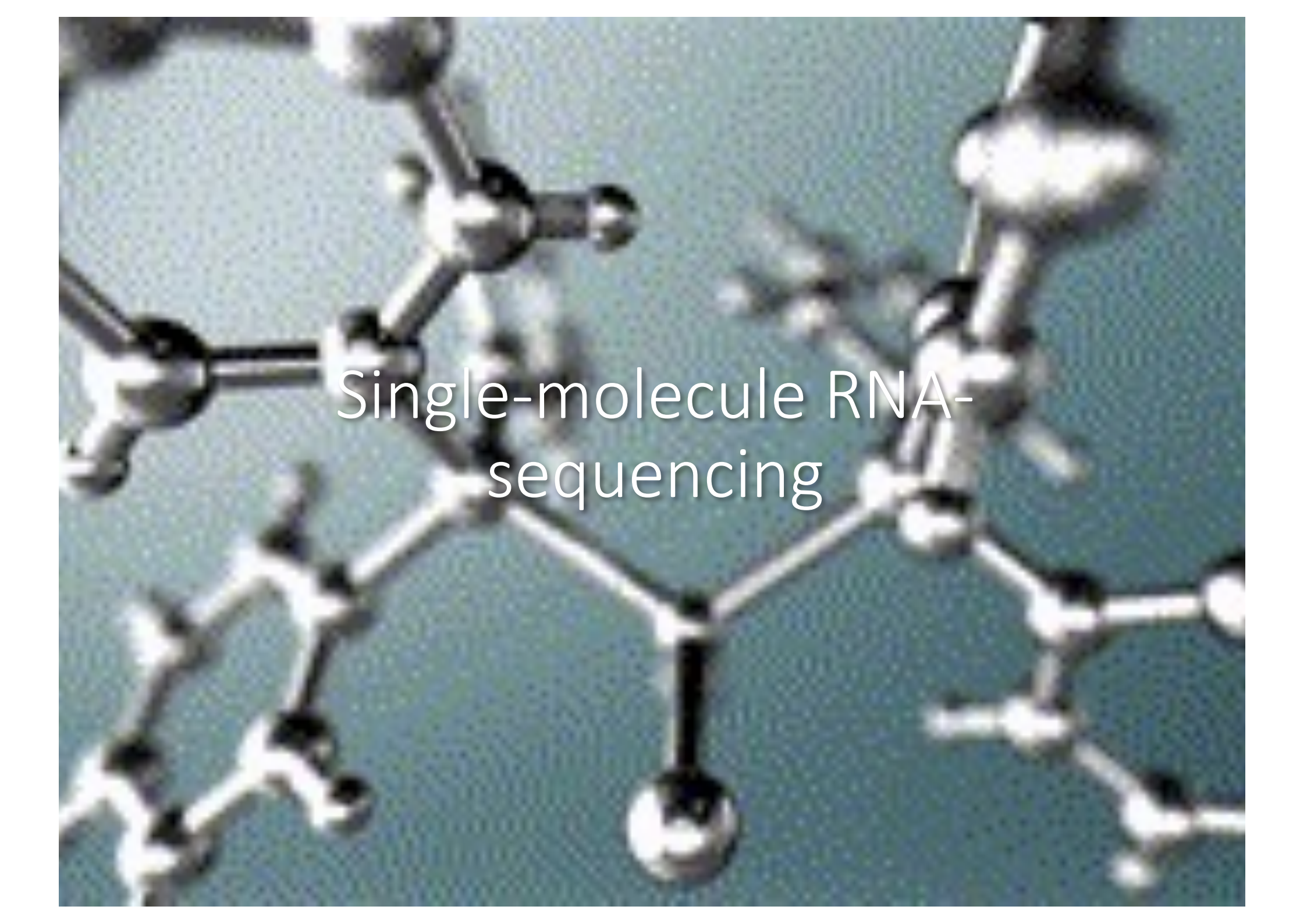
SMART-based vs Droplet-based

- SMART-based
 - Fresh tissue or frozen single cell suspension
 - One-cell at a time, lower throughput
 - Full length
 - Lower dropout
 - Transcripts / cell higher
 - Per cell transcription more accurate
- Droplet-based
 - Fresh tissue or frozen single cell suspension
 - All droplets together, higher throughput
 - Cell barcode
 - UMI for PCR bias correction
 - 3' (or 5') bias
 - Higher dropout
 - Transcripts / cell lower
 - Per cluster transcription more accurate

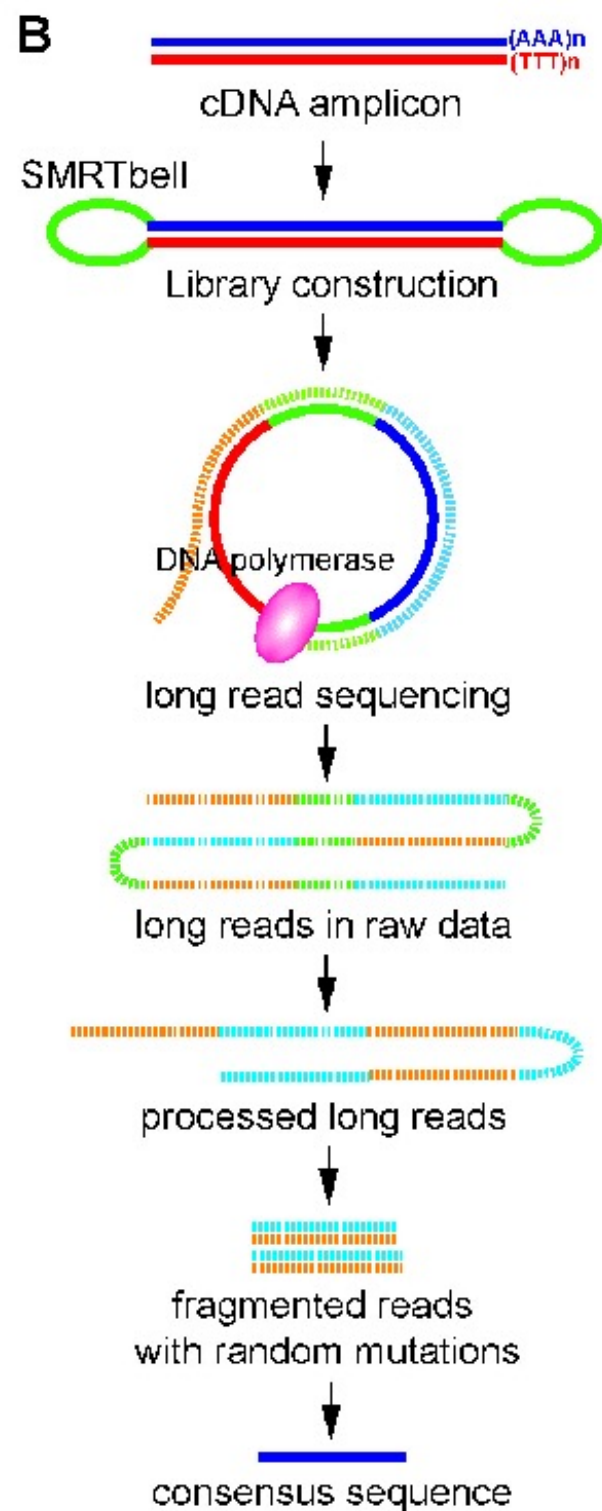
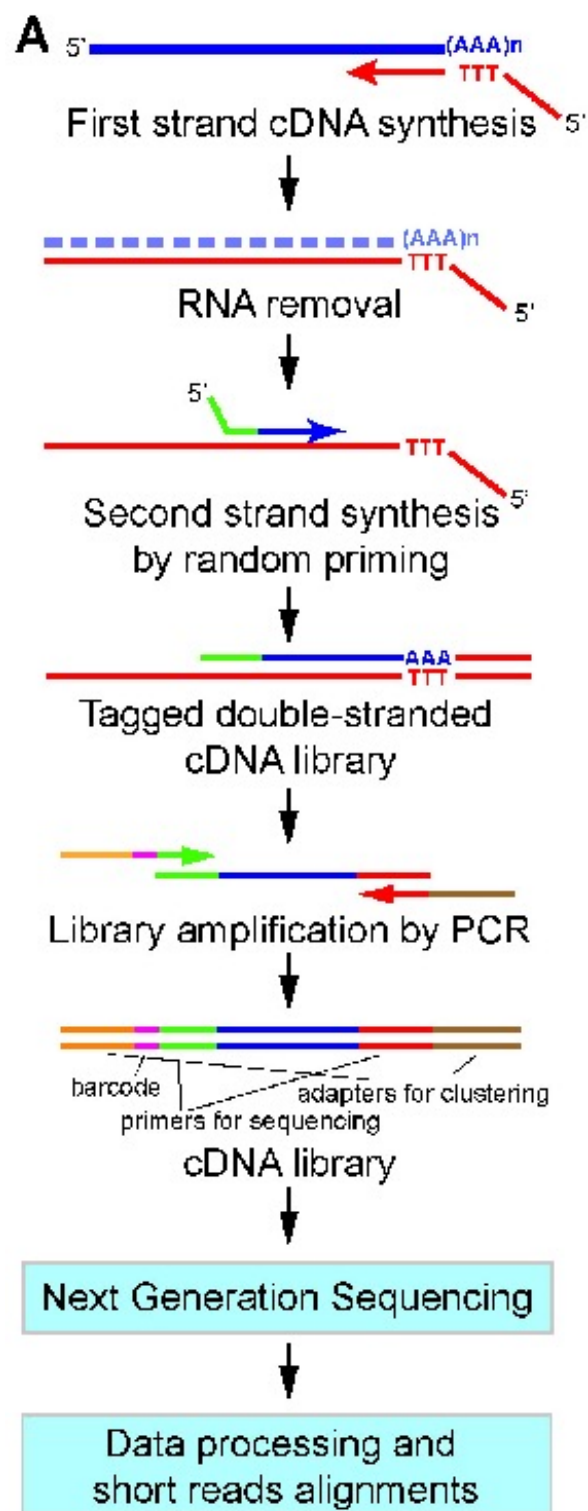
RNA FISH Validates Single-Cell RNA-Seq



Key single cell results can be validated using amplification-free methods.



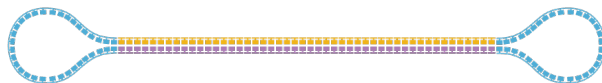
Single-molecule RNA-sequencing



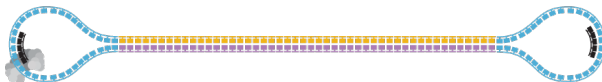
Start with high-quality double stranded DNA



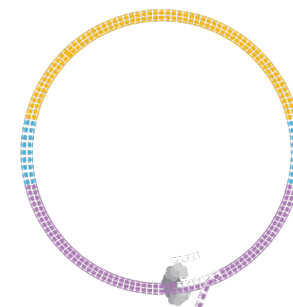
Ligate SMRTbell adapters and size select



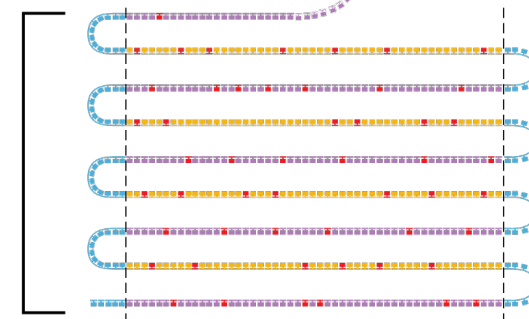
Anneal primers and bind DNA polymerase



Circularized DNA is sequenced in repeated passes



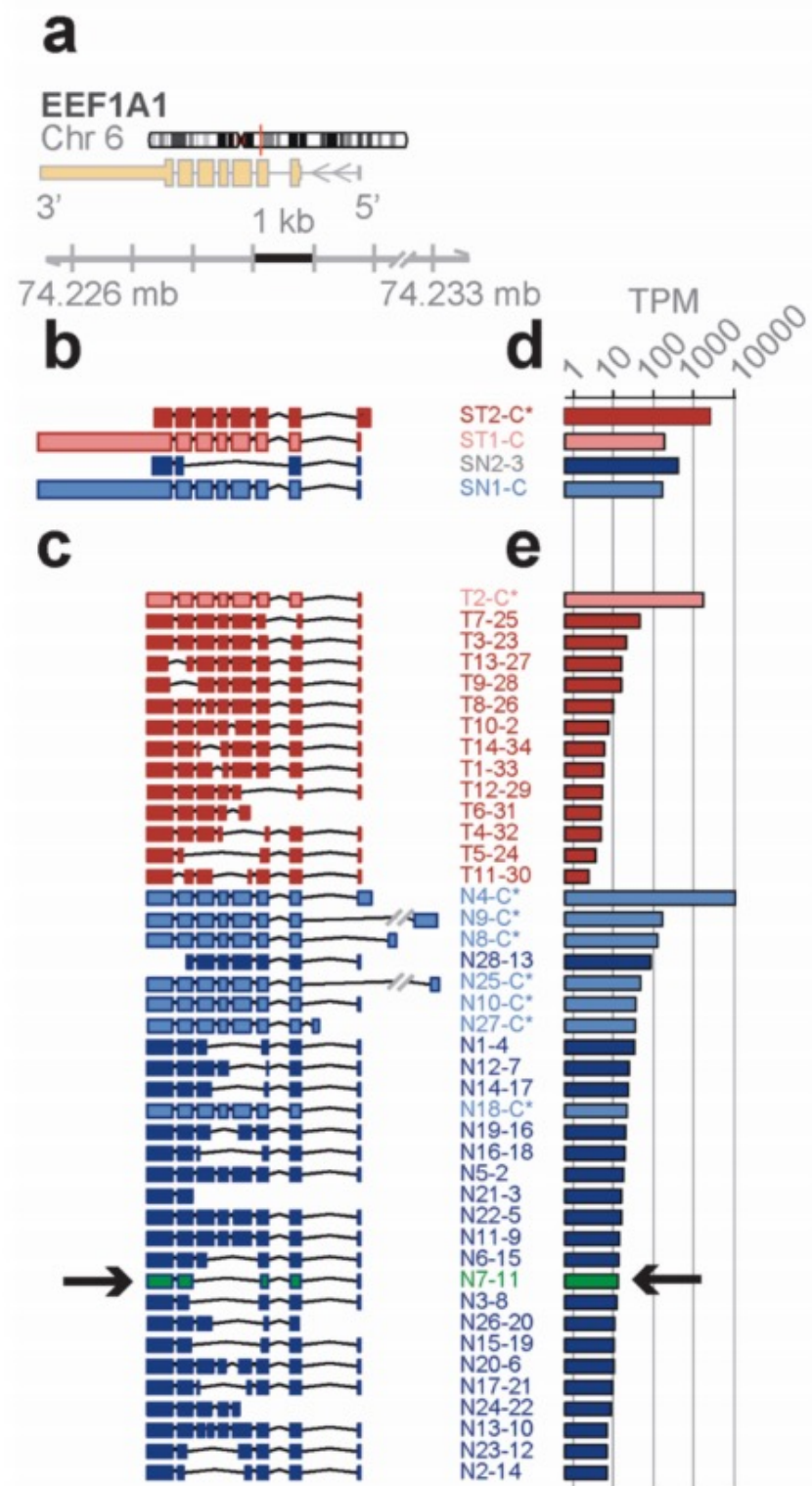
The polymerase reads are trimmed of adapters to yield subreads



Consensus is called from subreads

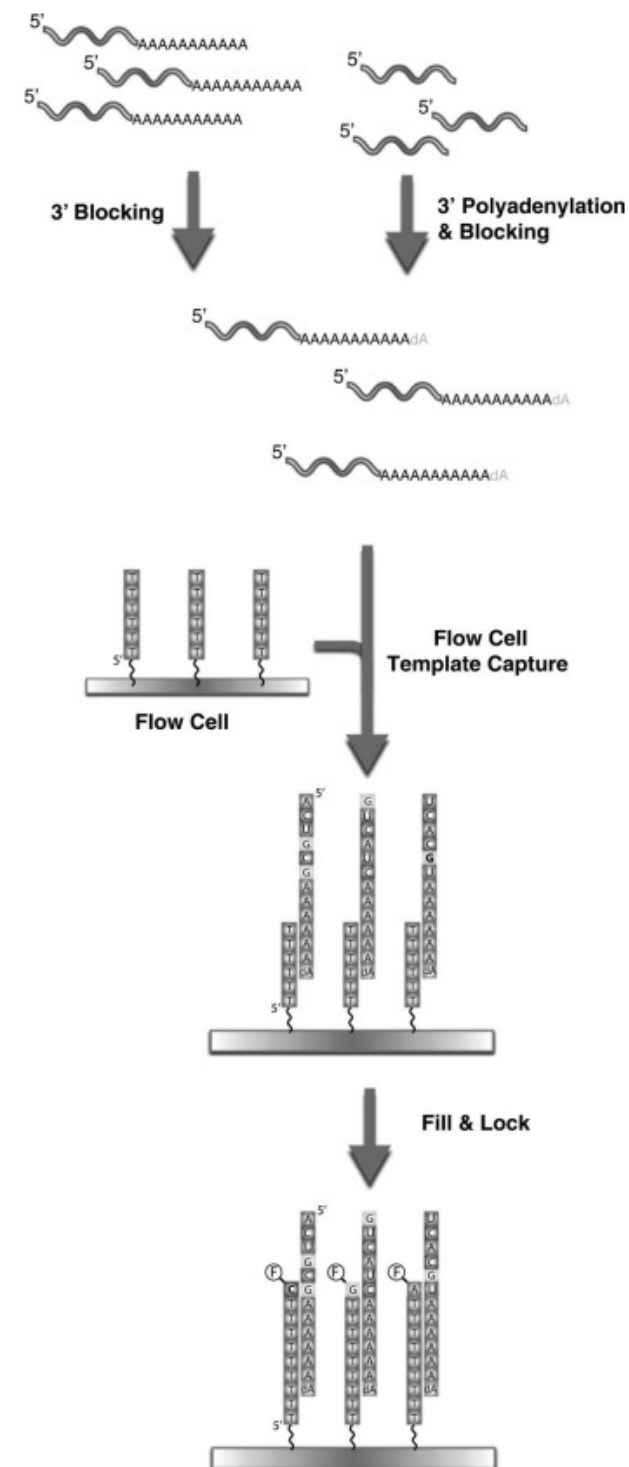


HiFi READ
(>99% accuracy)



Transcriptome Profiling Using Single-Molecule Direct RNA Sequencing

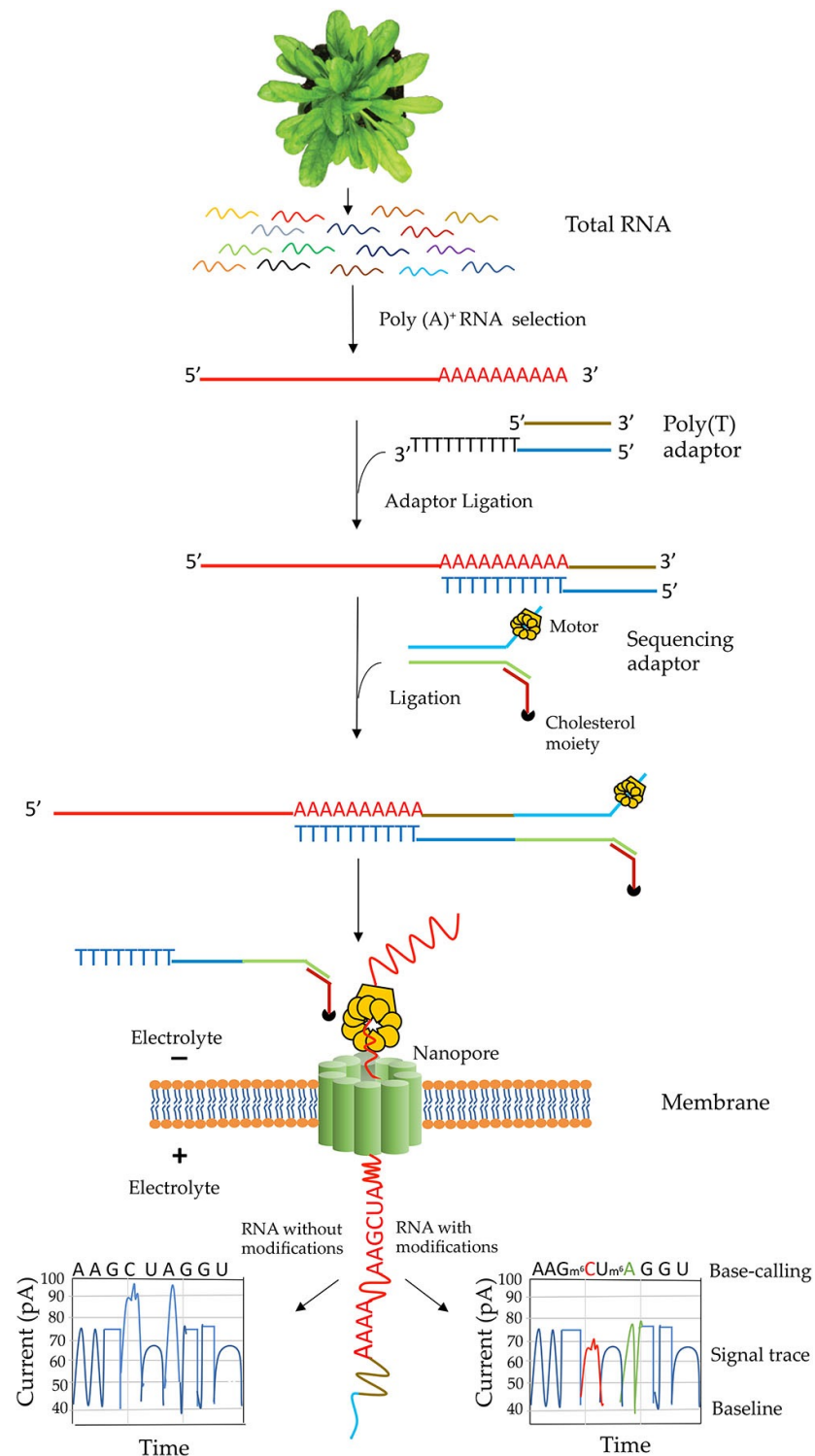
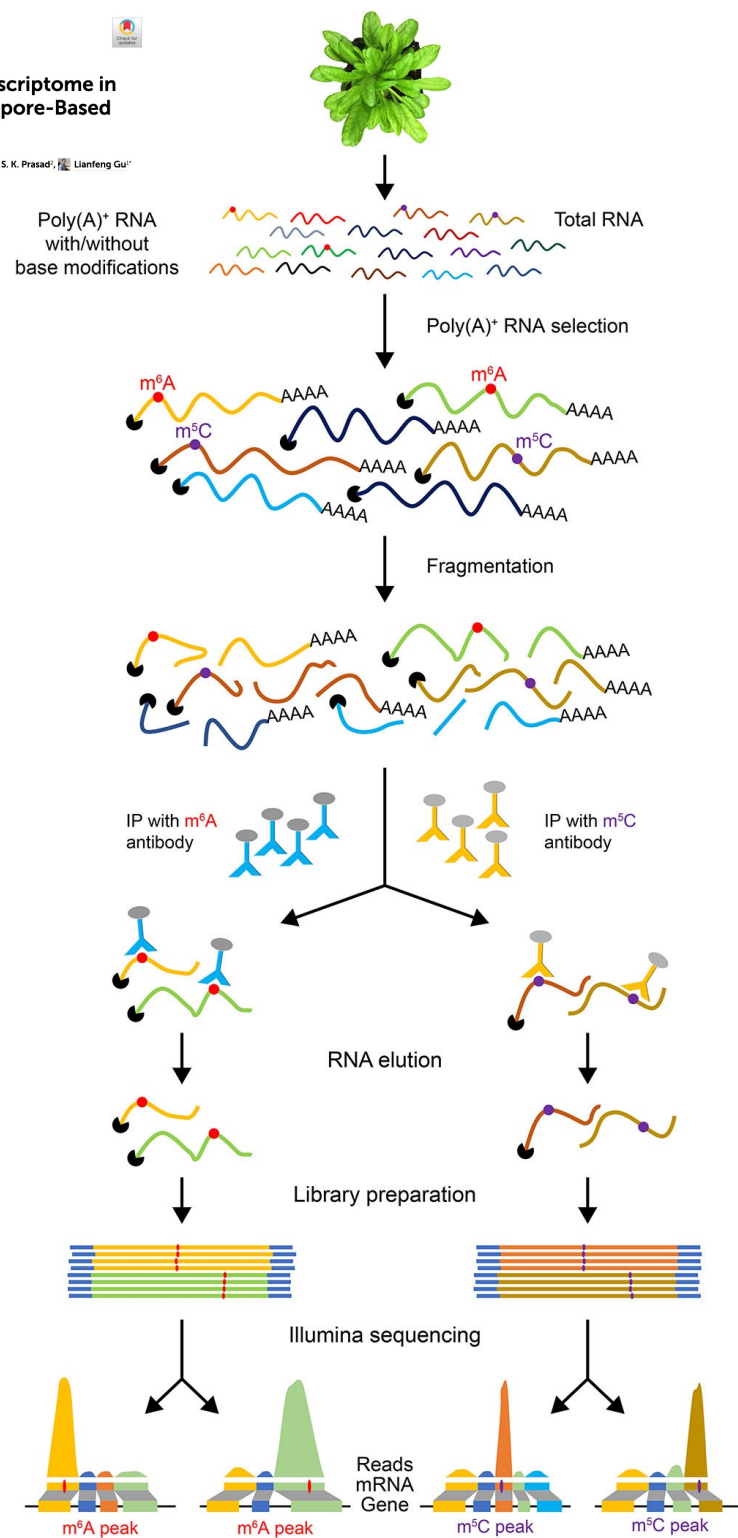
Fatih Ozsolak and Patrice M. Milos





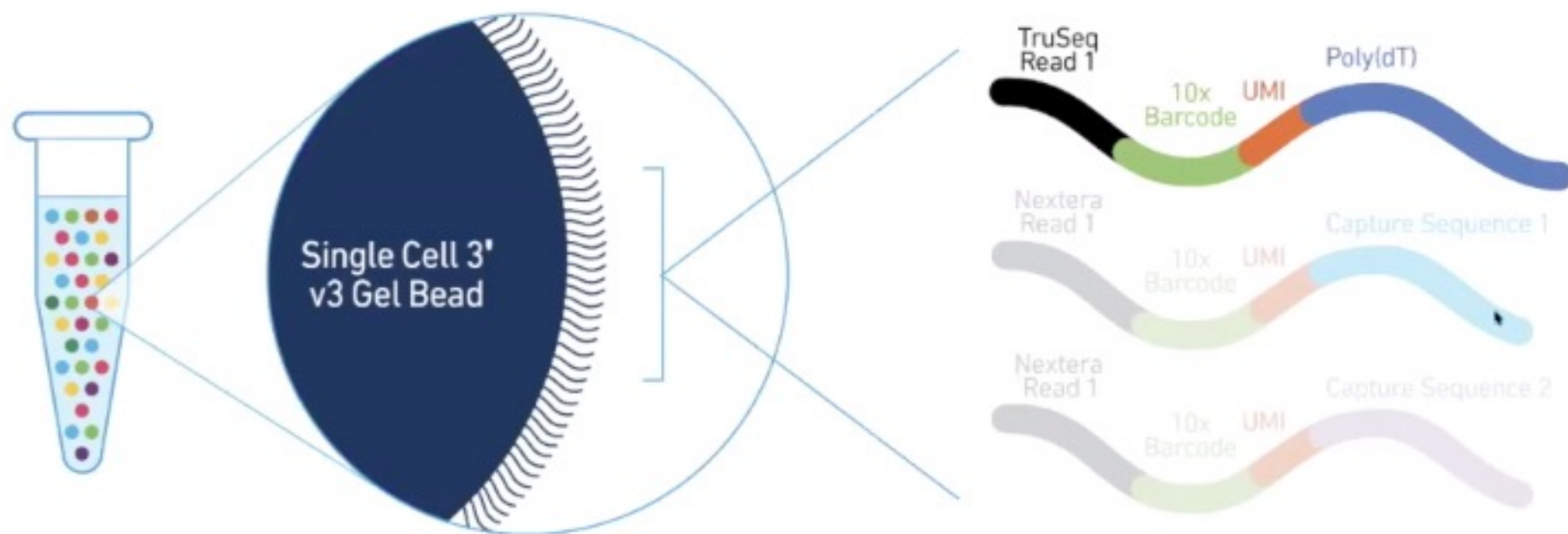
Analysis of Transcriptome and Epitranscriptome in Plants Using PacBio Iso-Seq and Nanopore-Based Direct RNA Sequencing

Liangzhen Zhao¹, Hangxiao Zhang¹, Markus V. Kohnen², Kasavajhala V. S. K. Prasad¹, Lianfeng Gu¹ and Anireddy S. N. Reddy^{1*}



Single Cell 3' v3.1 Gel Beads

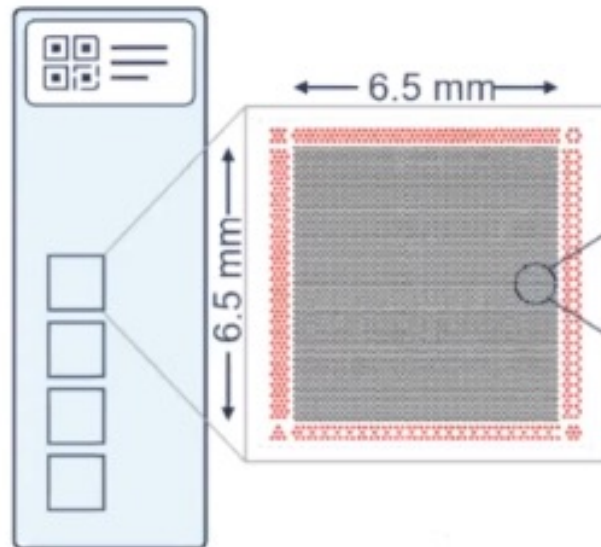
Feature Barcode technology enabled



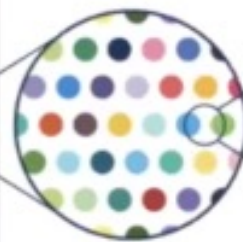
Whole Transcriptome Analysis with Spatial Resolution

Utilizing poly-A capture and unique spatial barcodes

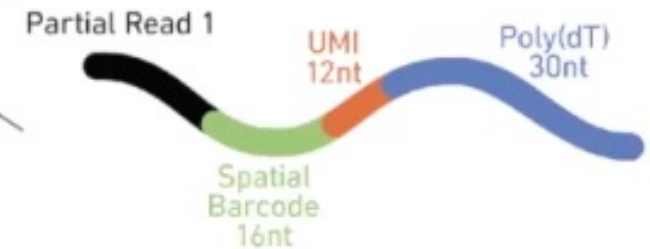
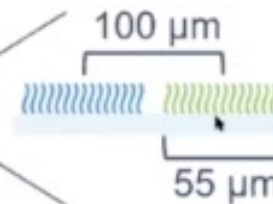
Visium Spatial Gene
Expression Slide



Capture Area with
~5000 Barcoded
Spots



Visium Gene
Expression Barcoded
Spots

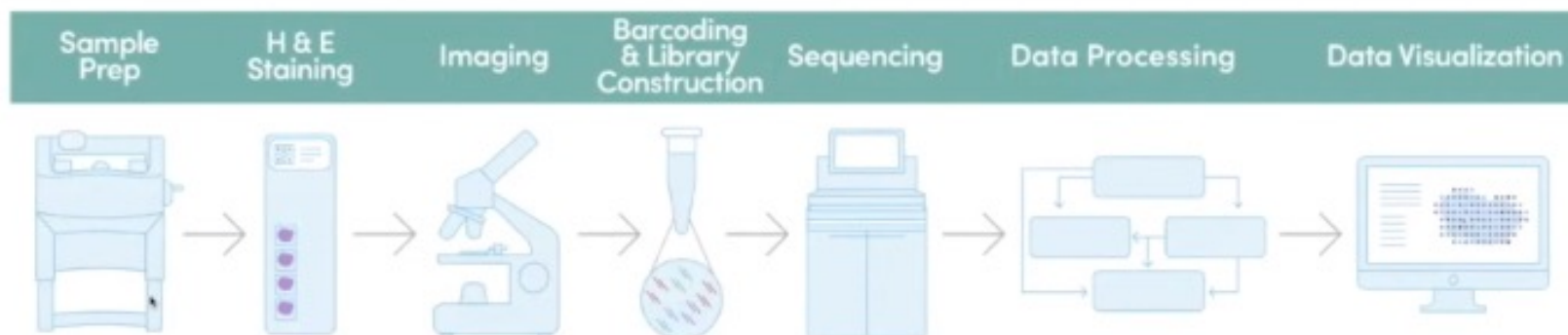


The Visium Spatial Gene Expression Workflow

Streamlined workflow with the choice of H&E or Immunofluorescence

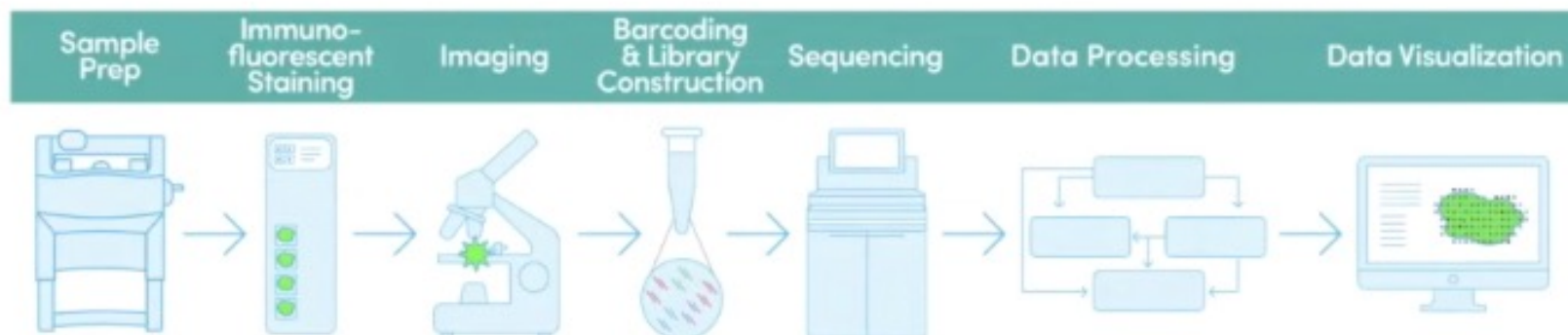
Visium with H&E Workflow

Provides whole transcriptome with morphological context



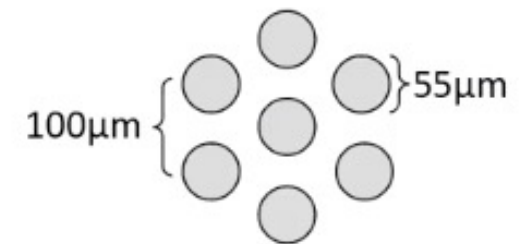
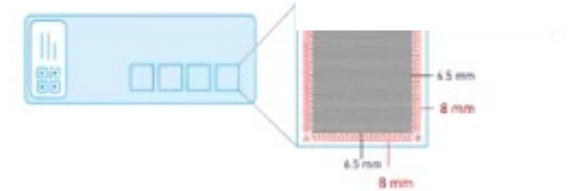
Visium with IF Workflow

Provides whole transcriptome with protein co-detection



Visium Platform

- Array based technique
- 6.5mm x 6.5mm area to put sample on
- 4992 spots arranged in hexagonal grid
- Spot specs:
 - Spot diameter : 55 μ m
 - Center to center distance : 100 μ m
- Each spot has millions of capture probes
 - spatial barcode
 - polyT sequence
 - captures polyadenylated mRNA
 - Full transcriptome(-ish)
- ~ 1-10 cells contribute to each spot
 - **NOTE** : Not single cell resolution!



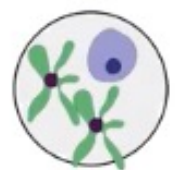
Spot 1



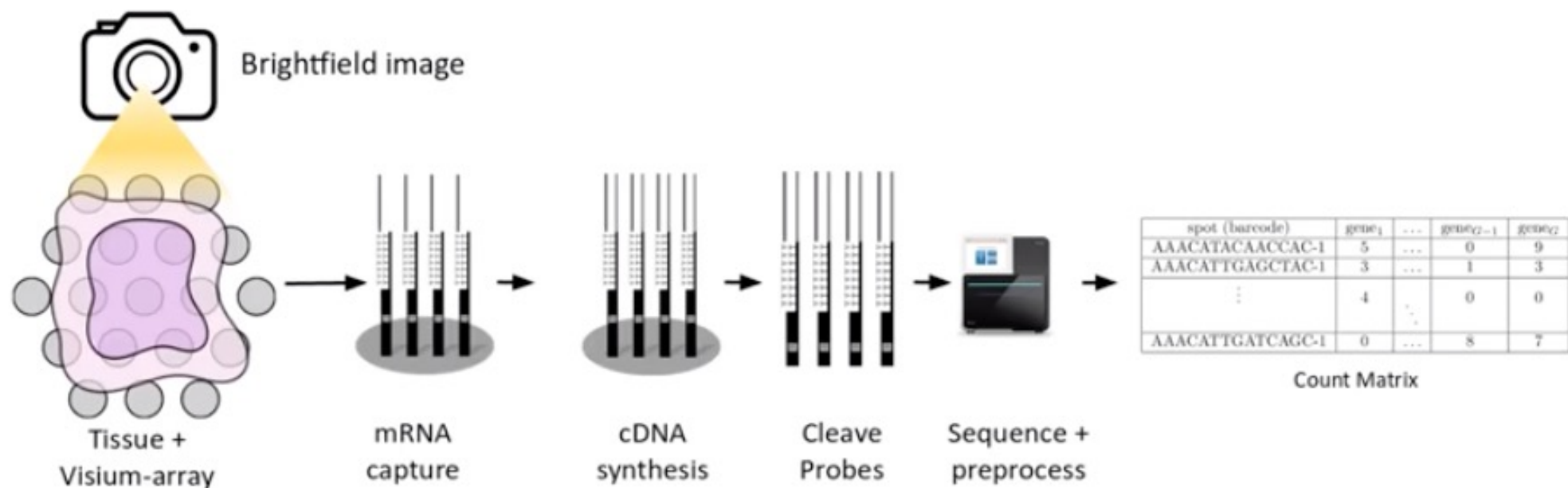
Spot 2



Spot 3



■■■■ The experimental workflow (in a nutshell)

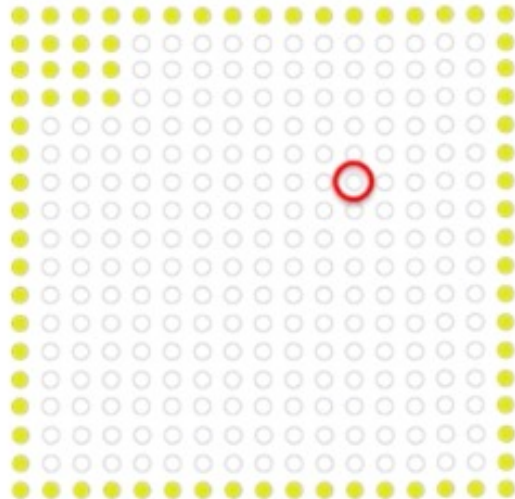


The ST Library Preparation Slide

Glass slide with array of capture probes

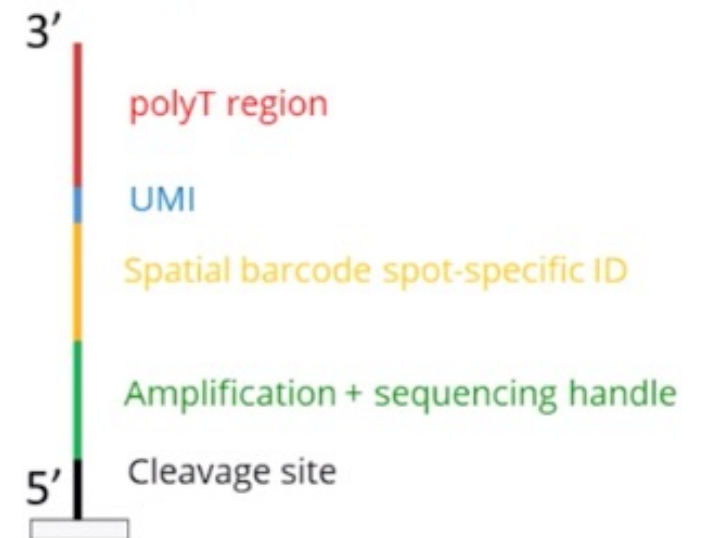


Sub-region with capture probes

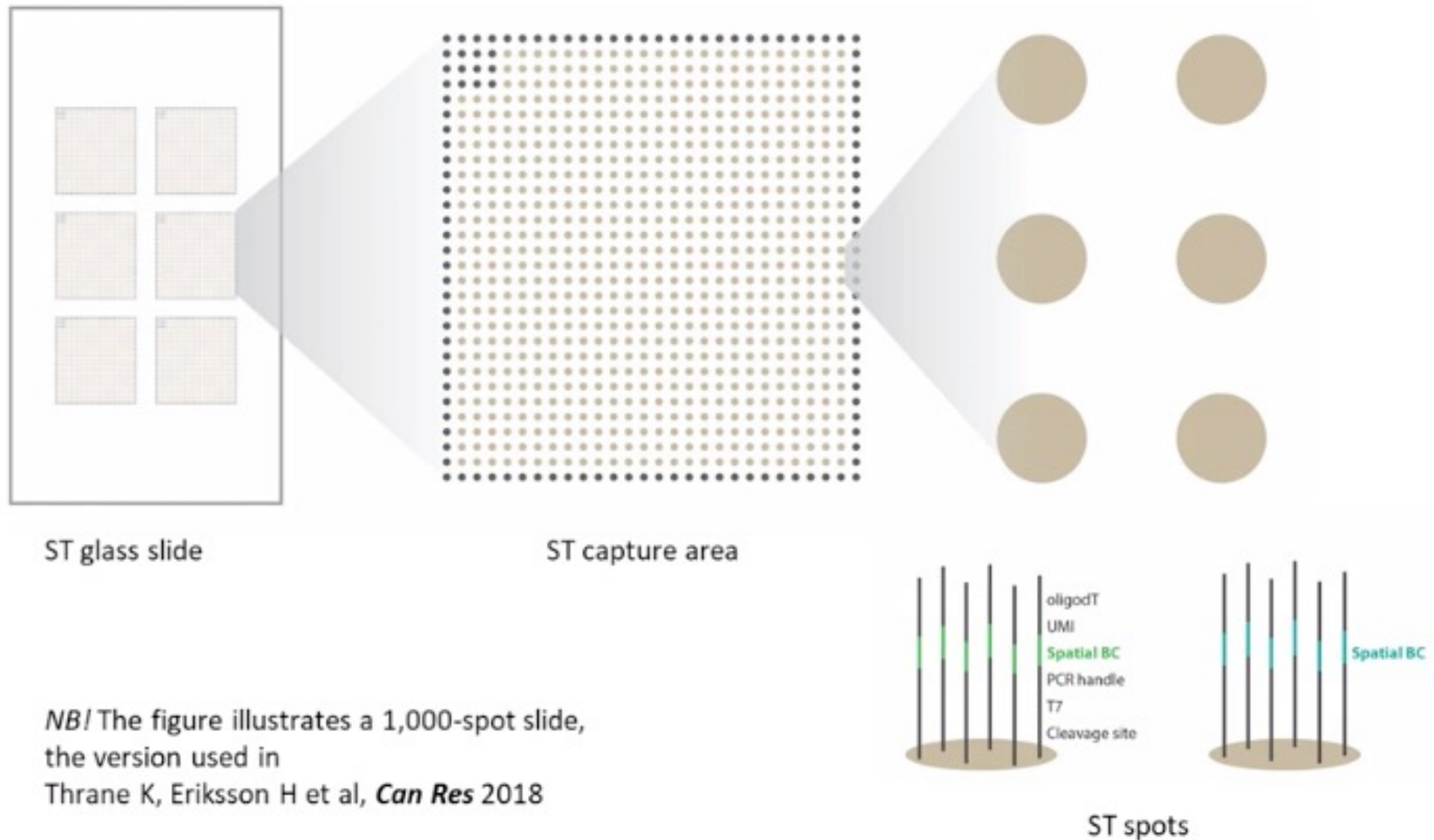


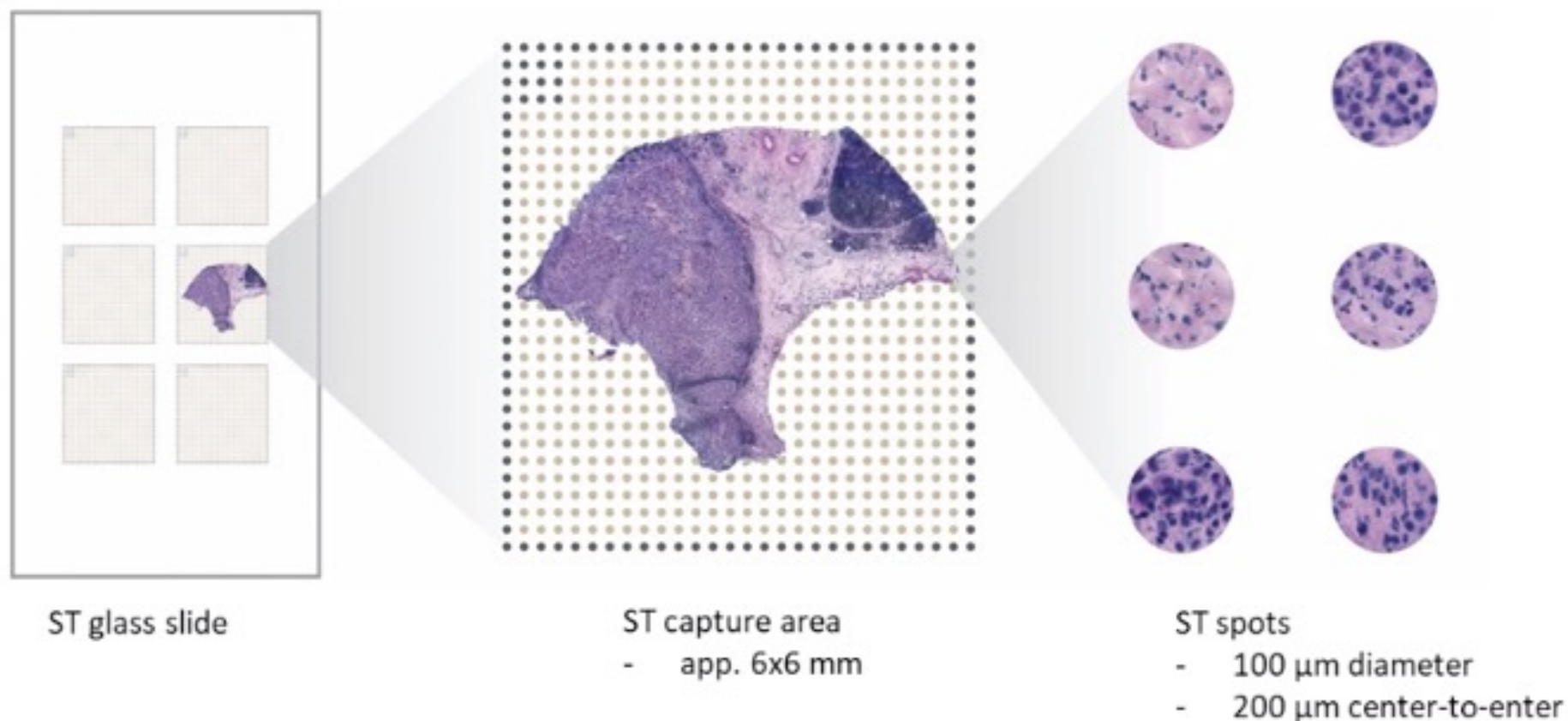
- 1007 spots in total
- 100 μm diameter
- Millions of probes /spot

Capture Probe



The ST slide



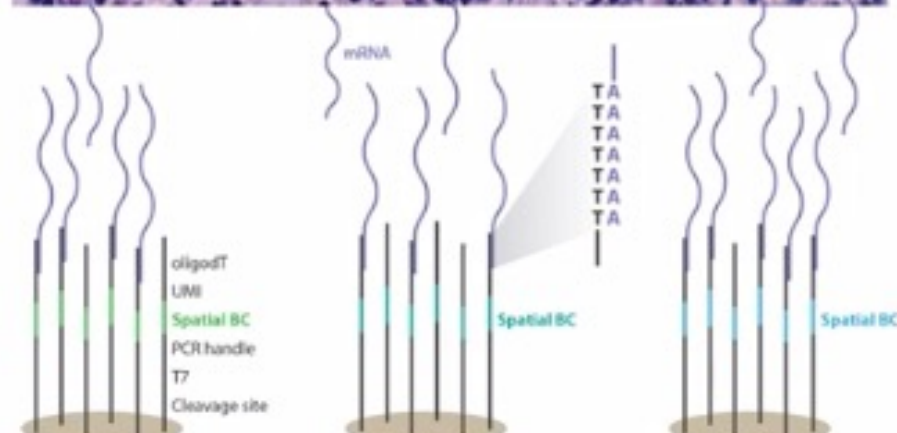


Slide version	No of spots	Spot diameter (μm)
ST - 1st gen.	1,000	100
ST - 2nd gen.	2,000	100
10x Visium	5,000	55

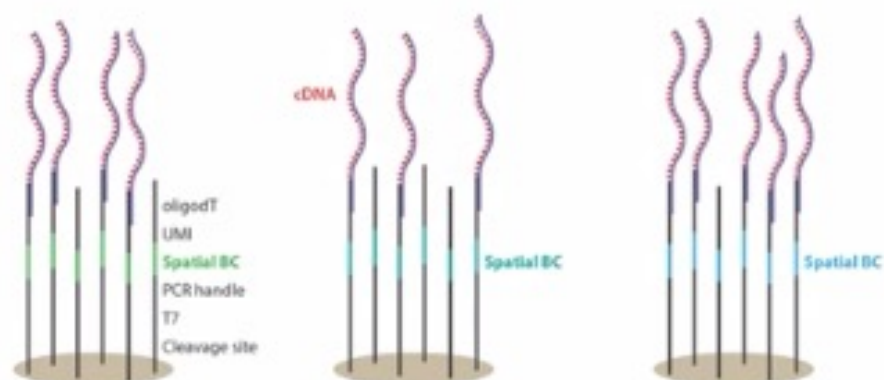
Increased
spot density
↓



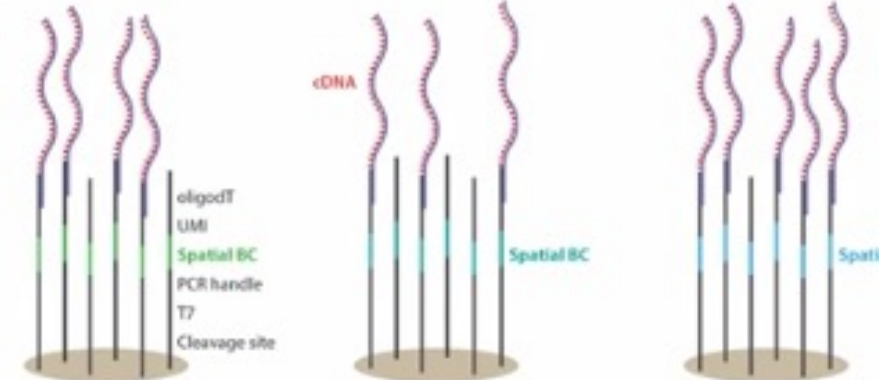
1) Tissue Fixation, H&E staining & Imaging



2) Tissue Permeabilization & mRNA Capture



3) Reverse Transcription

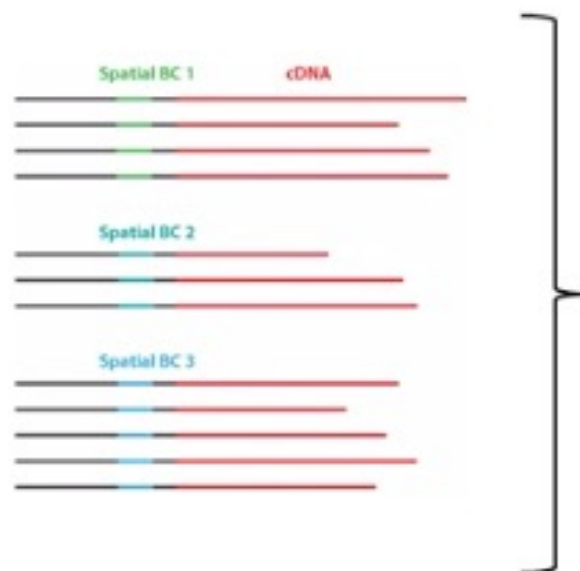


4) Tissue removal and cDNA cleavage

Library preparation & Sequencing



Slide cassette with rubber mask enables an individual reaction chamber for each capture area



Pooling of barcoded cDNA

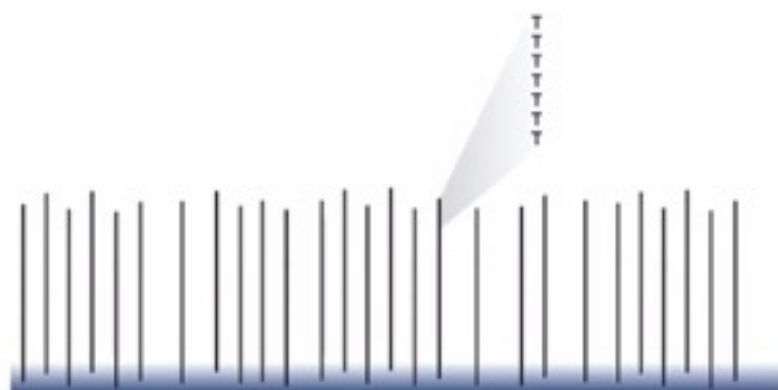


1 sample / capture area
ST automated library preparation:
Jemt A et al *Sci Rep* 2016

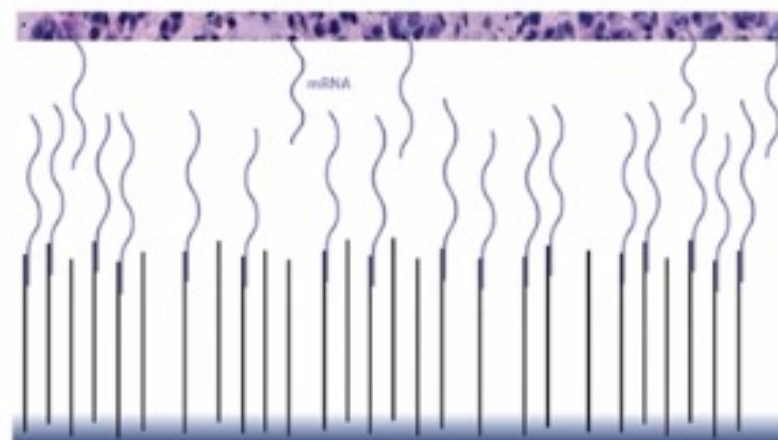


QC Slide for Tissue Optimization

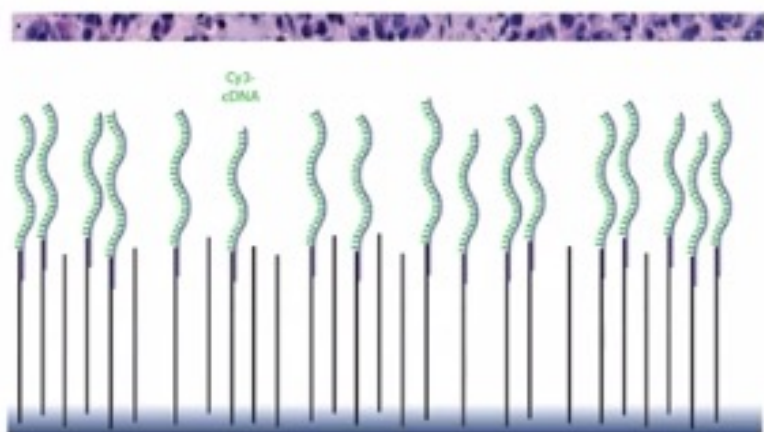
Different tissue types may require adaption of the protocol, especially permeabilization time.



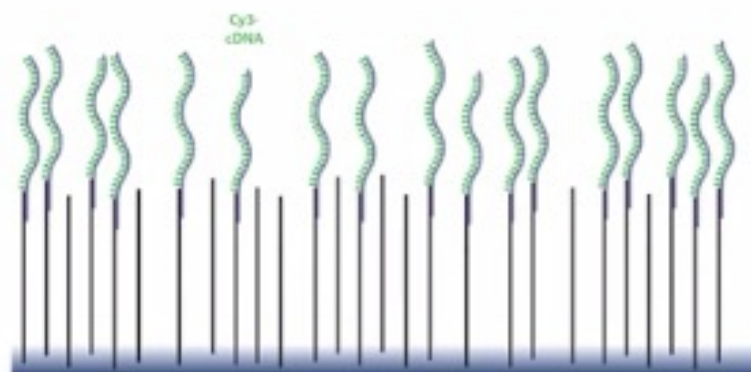
1) oligodT primers without spatial barcodes



2) Tissue Permeabilization & mRNA Capture



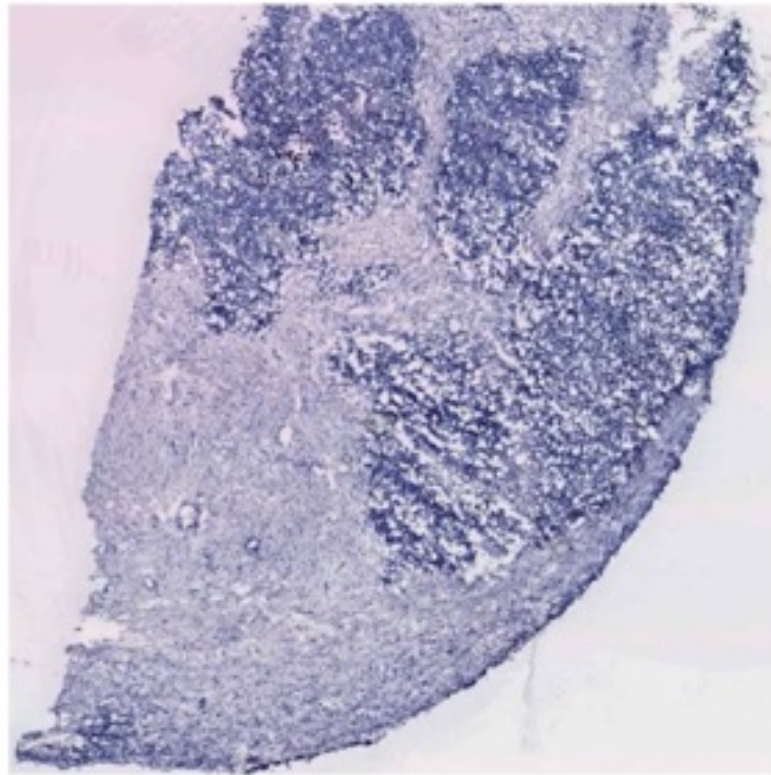
3) RT with fluorescently labeled nucleotides



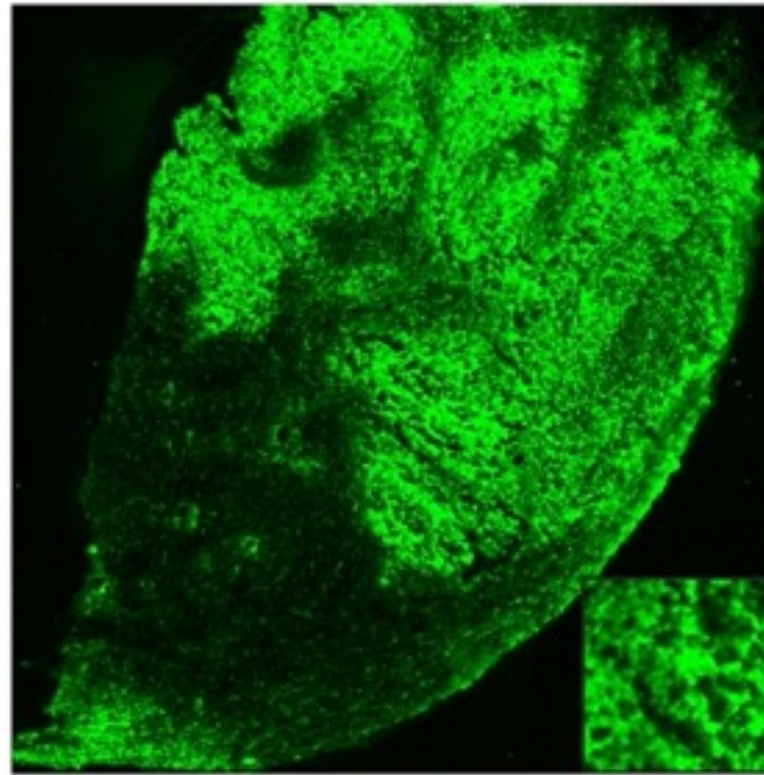
4) Tissue Removal & Fluorescence Imaging

cDNA “footprint” reflects tissue morphology

H&E image



Fluorescent cDNA



300µm

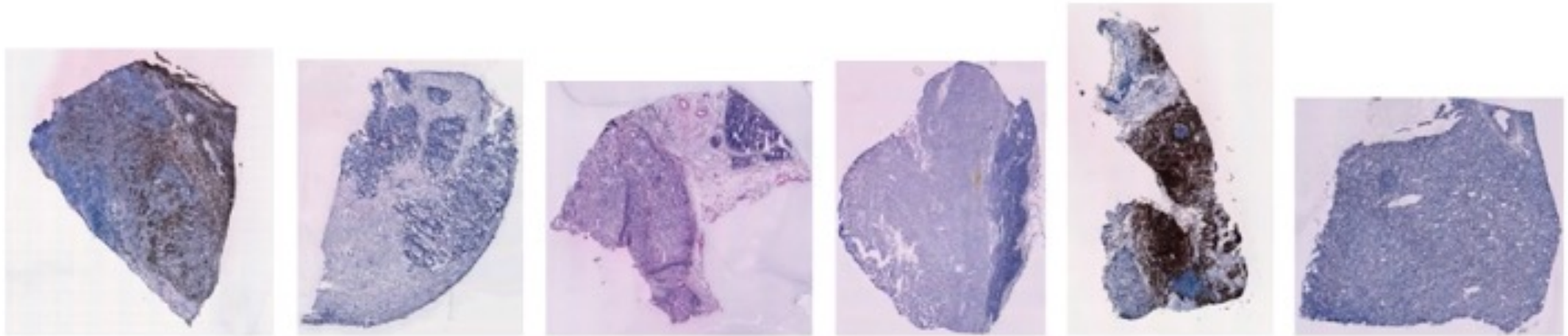
300µm

Optimized permeabilization time gives maximized mRNA capture without horizontal diffusion.

Additional parameters to take into account:

- Freezing procedure
- Section thickness
- Staining protocol
- RIN value

Melanoma Lymph Node Metastases



Tissue-specific concerns:

Differences in size and cell type composition

- Optimized permeabilization time for each sample

Will pigment in the sample inhibit RT?

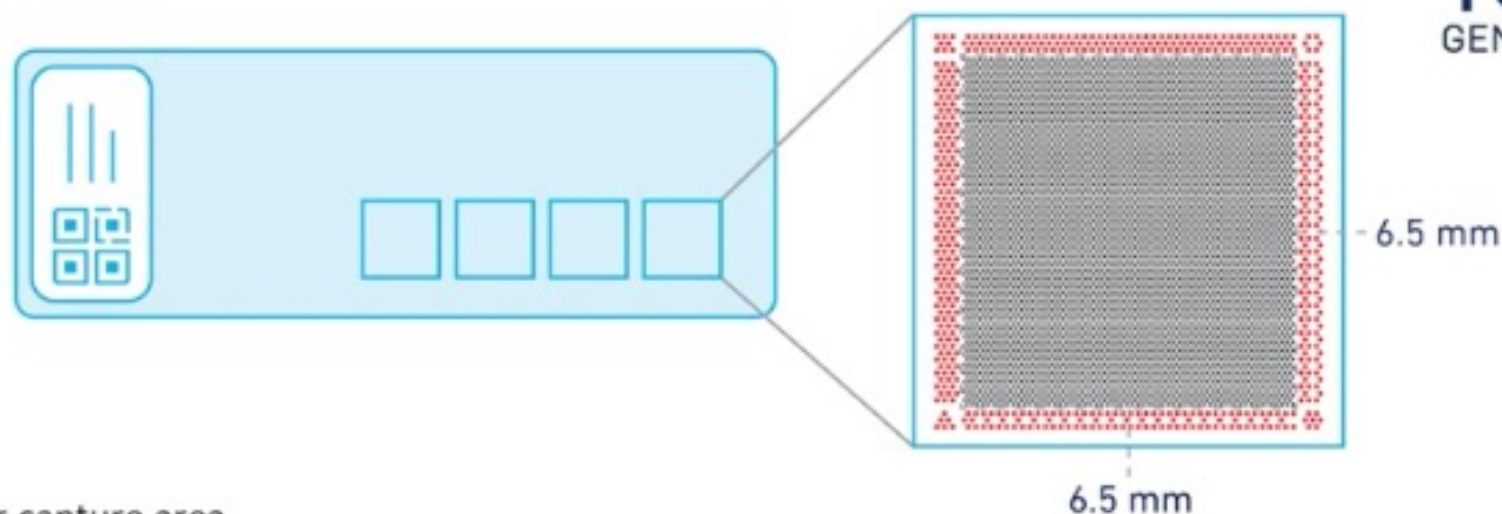
- Made tissue removal problematic but not necessarily RT
- Pigmented samples were not included in the first study

Sample age:

- The samples included in the study had been stored at -80°C for up to 20 years
- RIN values ~ 7

Visium Spatial Gene Expression Slide

10x
GENOMICS



Highlights

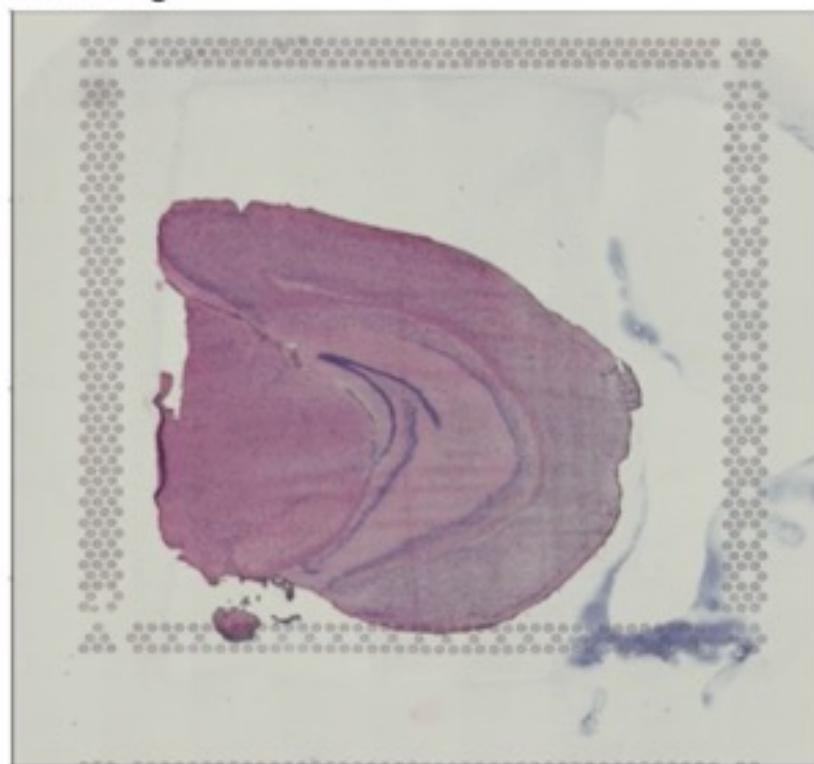
- 5000 spots per capture area
- 4 capture areas per slide
- Spot diameter of 55 μ m
- Center-to-center distance of 100 μ m
- 1-10 cells per spot, depending on tissue type
- Red border fiducials are used to align sequencing data to image data

Slide version	No of spots	Spot diameter (μ m)
ST - 1st gen.	1,000	100
ST - 2nd gen.	2,000	100
10x Visium	5,000	55

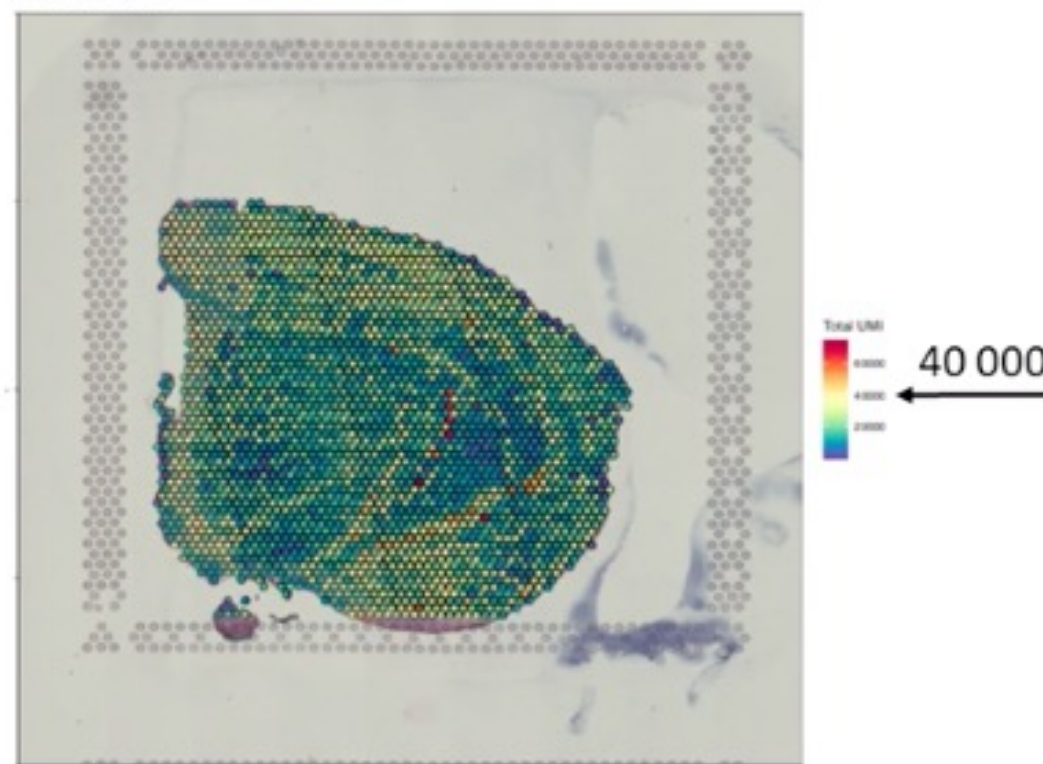
Increased
spot density
↓

Visium Gene Expression data of Mouse hippocampus

H&E image



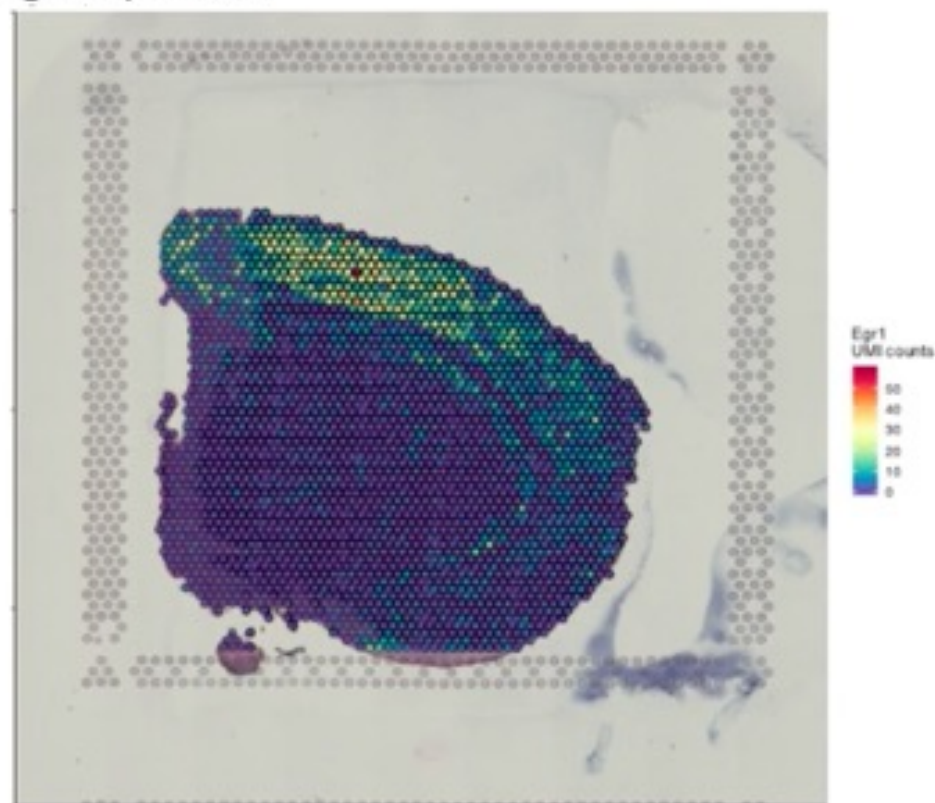
Total UMI



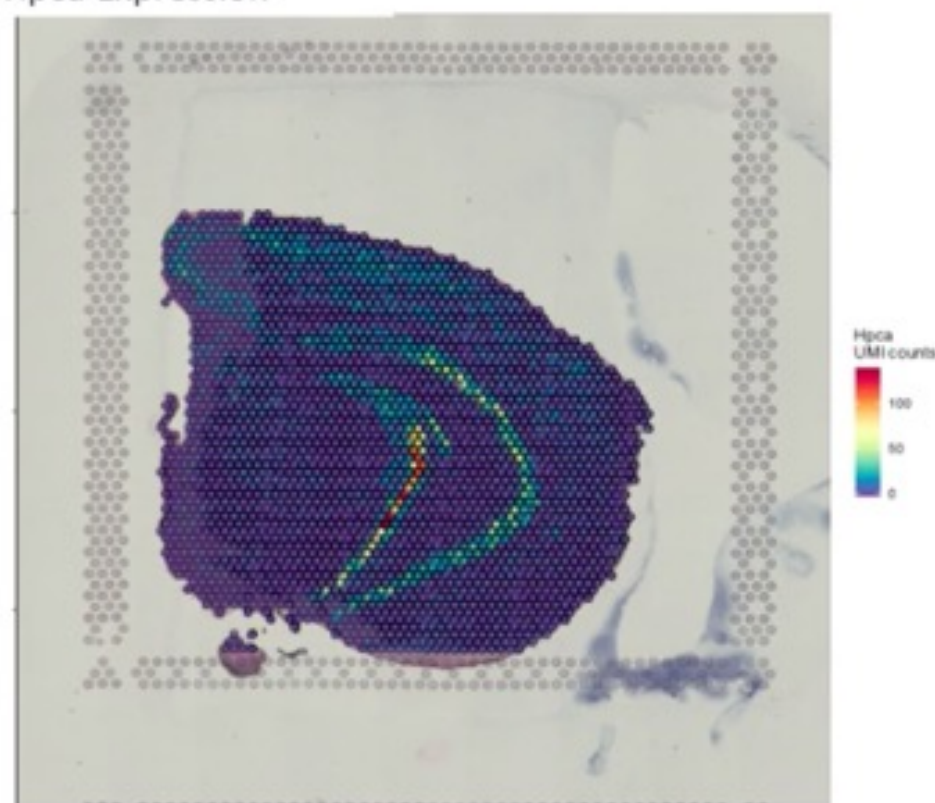
H&E staining image and total UMIs

Total spots covered by tissue was exactly 2500 (out of 5000).

Egr1 Expression

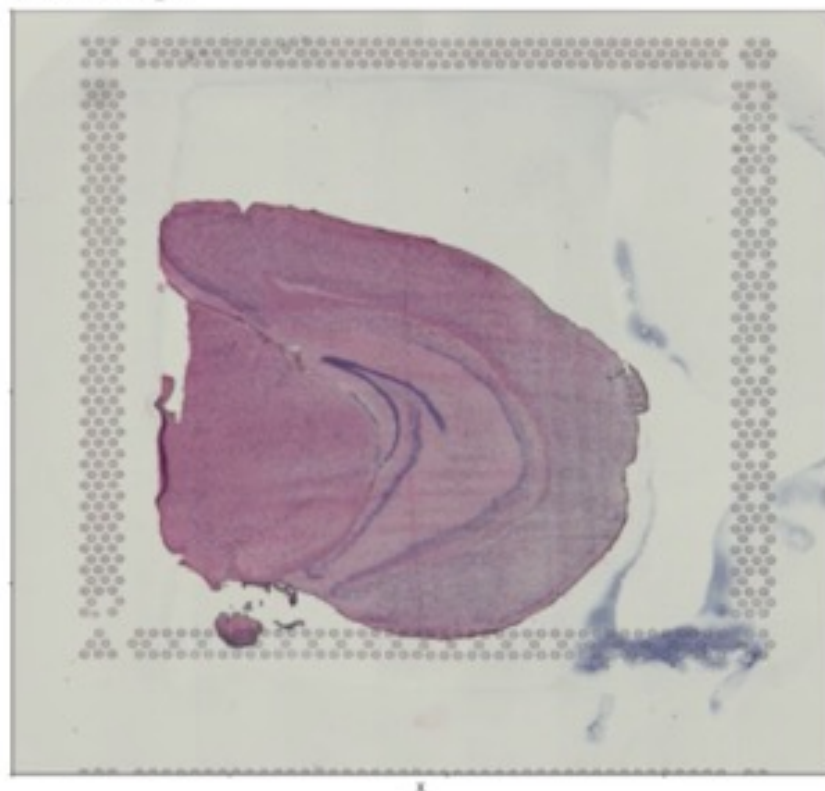


Hpca Expression

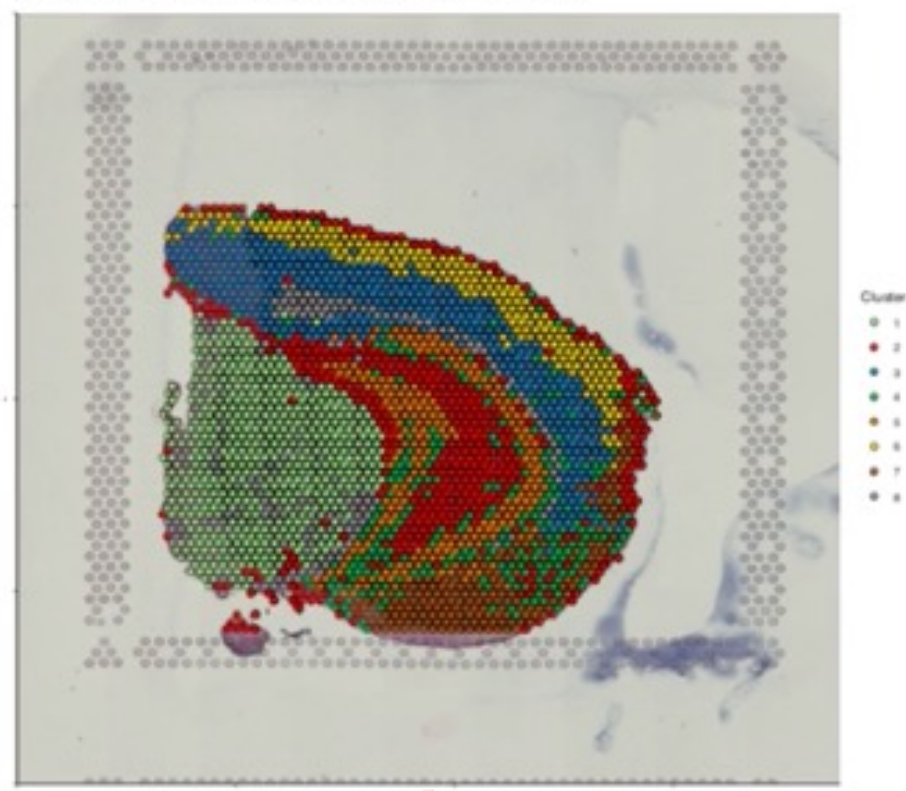


Spatially distributed expression of Egr1 and Hpca

H&E image



Spatial Gene Expression Clustering



H&E staining image and spatial gene expression clustering

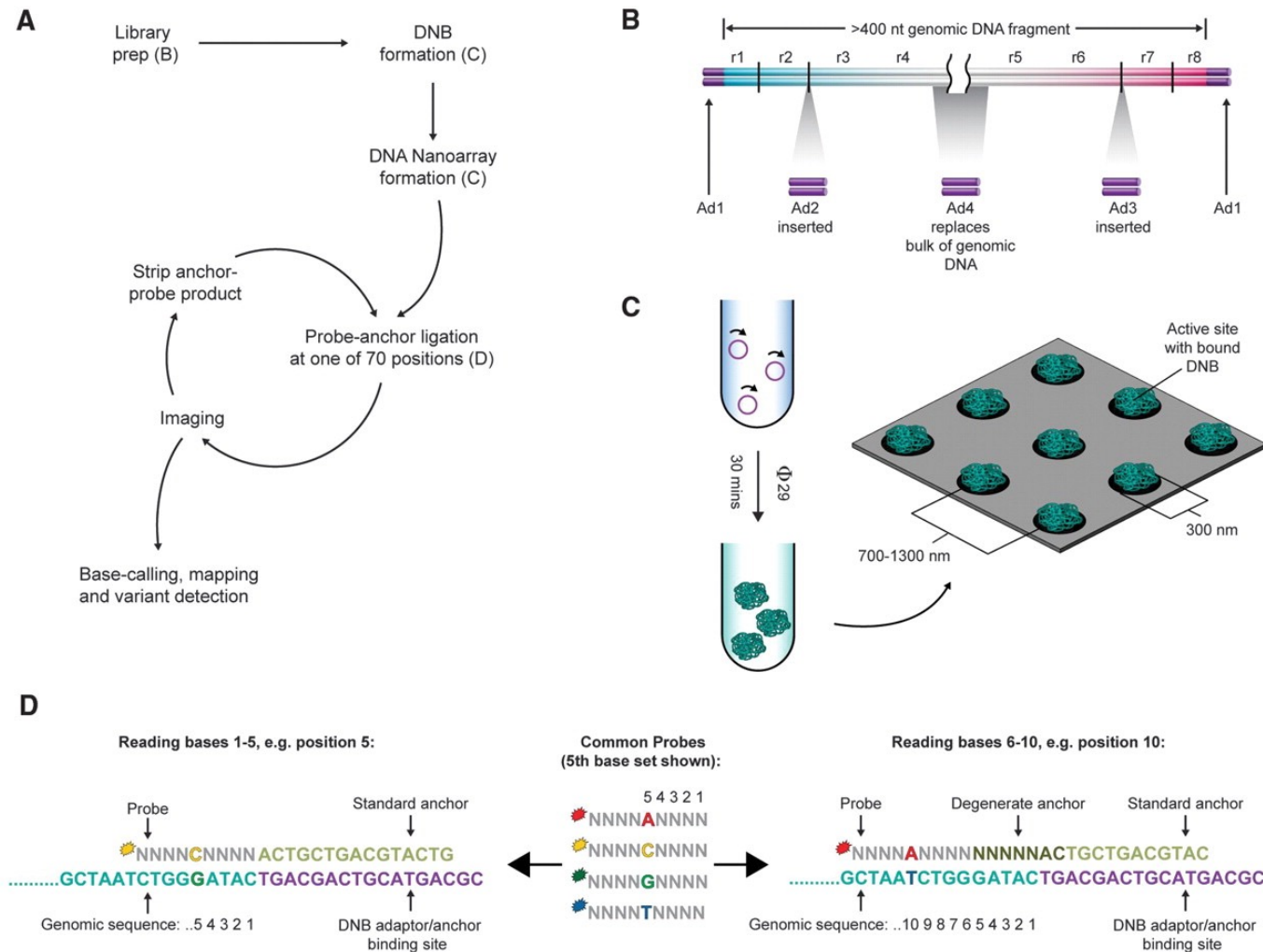
DNA nanoball sequencing

- **DNA nanoball sequencing** is a [high throughput sequencing](#) technology that is used to determine the entire [genomic sequence](#) of an organism. The method uses [rolling circle replication](#) to amplify small fragments of genomic DNA into *DNA nanoballs*.

Human Genome Sequencing Using Unchained Base Reads on Self-Assembling DNA Nanoarrays

RADOJE DRMANAC, ANDREW B. SPARKS, MATTHEW J. CALLOW, AARON L. HALPERN, NORMAN L. BURNS, BAHRAM G. KERMANI, PAOLO CARNEVALI, IGOR NAZARENKO,

GEOFFREY B. NILSEN [1, 2], AND CLIFFORD A. REID  [+55 authors](#) [Authors Info & Affiliations](#)



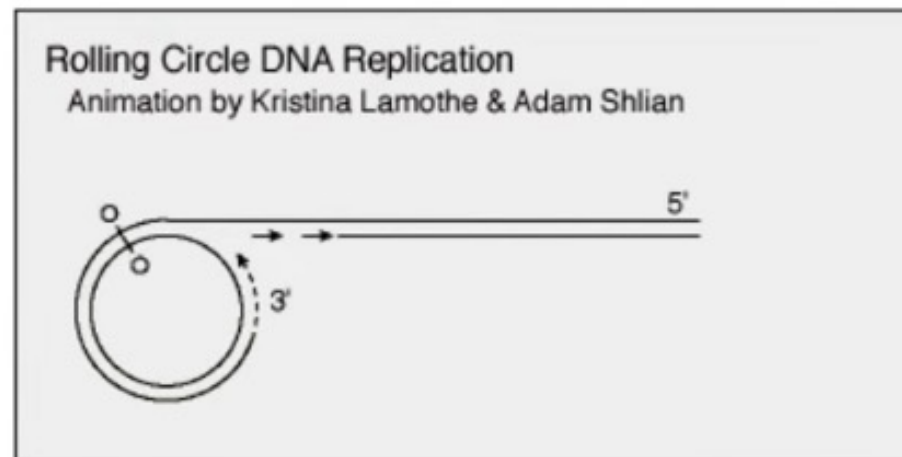
What is nanoball sequencing?

- A high throughput, 3rd generation sequencing technology
- Utilizes combinatorial Probe-Anchor Synthesis (cPAS) and DNA Nanoball (DNB) Technology



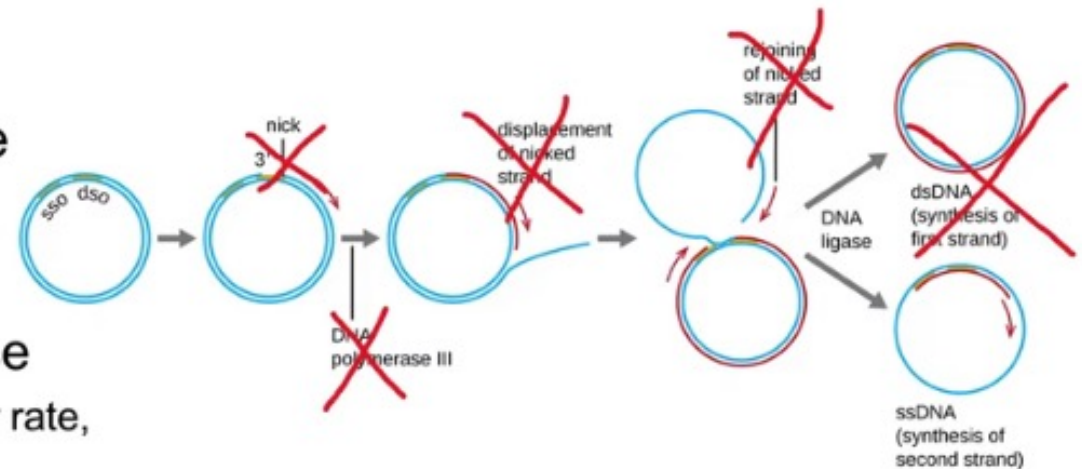
Nanoball Technology is a **modified version** of **rolling circle replication**

Rolling circle replication (bacterial replication) - process of unidirectional nucleic acid replication that can rapidly synthesize multiple copies of circular molecules of DNA or RNA.



Modified Rolling Circle Replication

- Does not create circular dsDNA
- No initiator proteins or nickases are present
- No displaced DNA, no cleavage
- Uses Phi 29 ($\Phi 29$) DNA polymerase
 - high processivity, high fidelity, low error rate, generates long fragments
 - From *Bacillus subtilis*
- Without nickases, polymerase forms long concatemers
 - Concatemers = continuous DNA molecule with multiple head-to-tail copies of the same DNA fragment



Information source: <https://www.nature.com/articles/nbt0110-43>

Image source: https://commons.wikimedia.org/wiki/File:OSC_Microbio_11_02_RollCircle.jpg

DNA Nanoball Library Preparation

1- Isolate Genomic DNA* (1 μ g, >23 kb is standard)

2 - Fragmentation of Sample

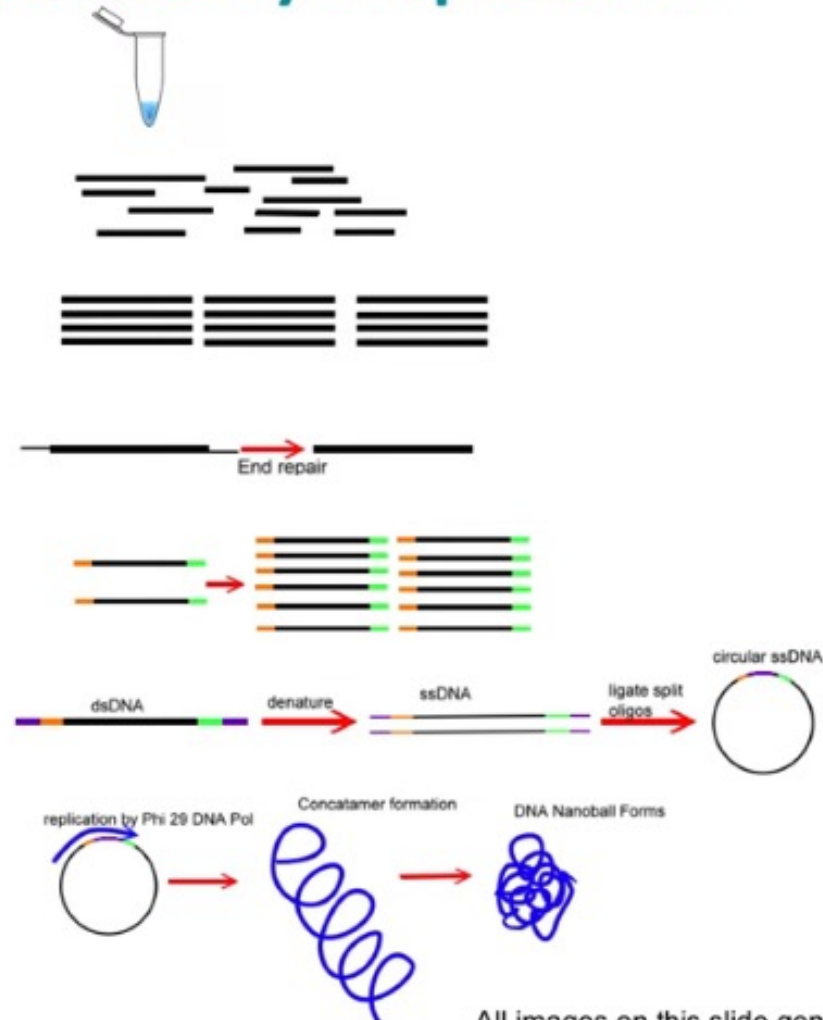
3 - Size Selection
(range: 50-800 bp; 100-300 bp optimal)

4 - End Repair

5 - Attach Adapter Sequences & PCR Amplify

6 - Attach Split Oligo, Denature DNA Strands, and
Ligate to form circular ssDNA

7 - Modified Rolling Circular Replication to form
DNBs

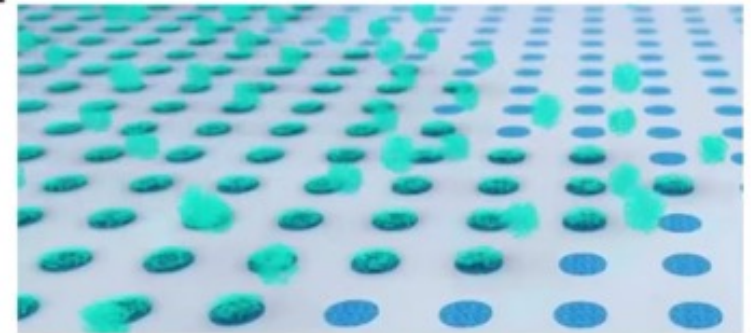


Information source:
<https://www.nature.com/articles/nbt0110-43>

All images on this slide generated by L. St. Clair

Loading Onto Flow Cell/Sequencing

- Once Nanoballs are formed they are loaded onto a patterned-array flow cell
- Flow cell = silicon wafer coated with silicon dioxide, titanium, hexamethyldisilane, and photoresist material
- Each Nanoball will selectively bind 1 positively-charged aminosilane
- Sequence data is determined stepwise, like Illumina:
 - dNTP incorporated by polymerase
 - Each dNTP is conjugated to a particular Fluor, and contains a terminator blocking additional extension
 - Un-incorporated dNTPs are washed
 - Image is captured
 - Dye and terminator cleaved
 - Repeat until sequencing is complete



Nanoballs loading onto pattern array.

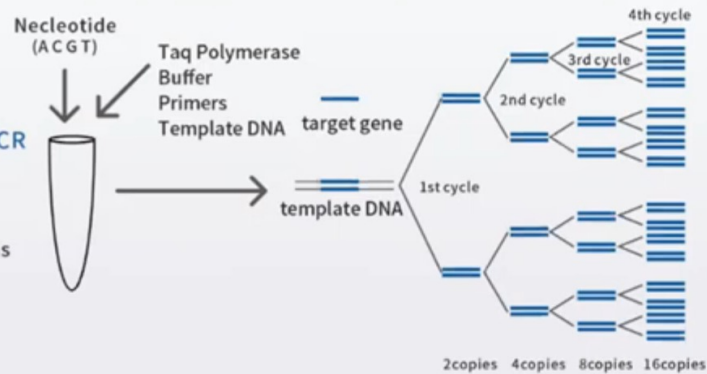
Image Captured from:
<https://www.youtube.com/watch?v=zjEQPGDx-J4>

MGI Sequencing Technology – DNBSEQ™

PCR Exponential Amplification

NGS problems
introduced by PCR

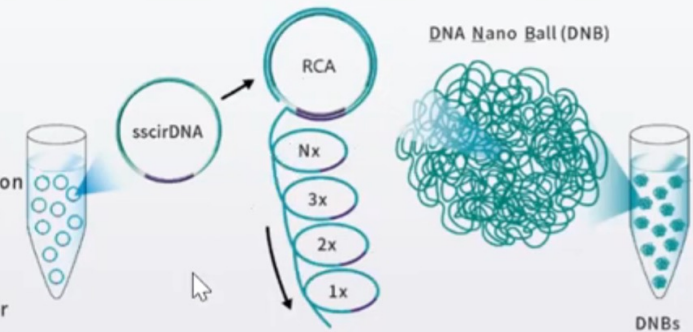
1. False SNPs
2. False InDels
3. Coverage/GC bias
4. Index hopping



RCA: MGI DNBSEQ™ Technology

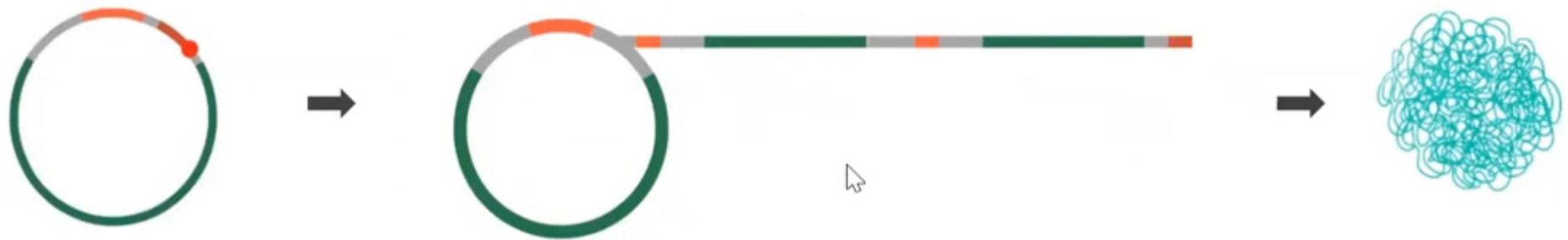
MGI DNA
nanoballs

1. Linear amplification
2. Low amplification bias
3. No accumulated amplification error
4. Low index hopping



The process of RCA reaction

1. Primer annealing
2. Extension
3. Displacement of DNA strand
4. Continue extension



RCA Advantages

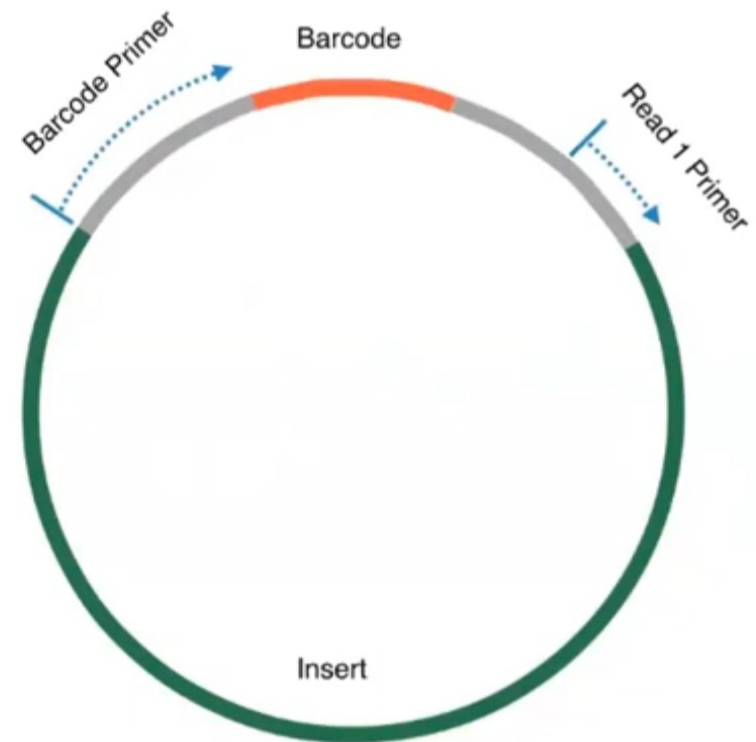
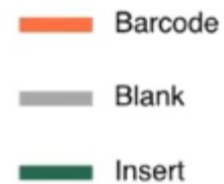
- Based on the original template
- Linear amplification
- Low amplification bias
- No error accumulation

★ **More accurate data**
> 99.9% SNP Precision/Sensitivity
> 99% Indel Precision/Sensitivity

*NA12878 library by using MGIEasy PCR-Free Library Preparation Kit

Single stranded circularized DNA structure

Insert and barcode sequencing
primer binding sites on ssCirDNA



Why DNA Nanoball?

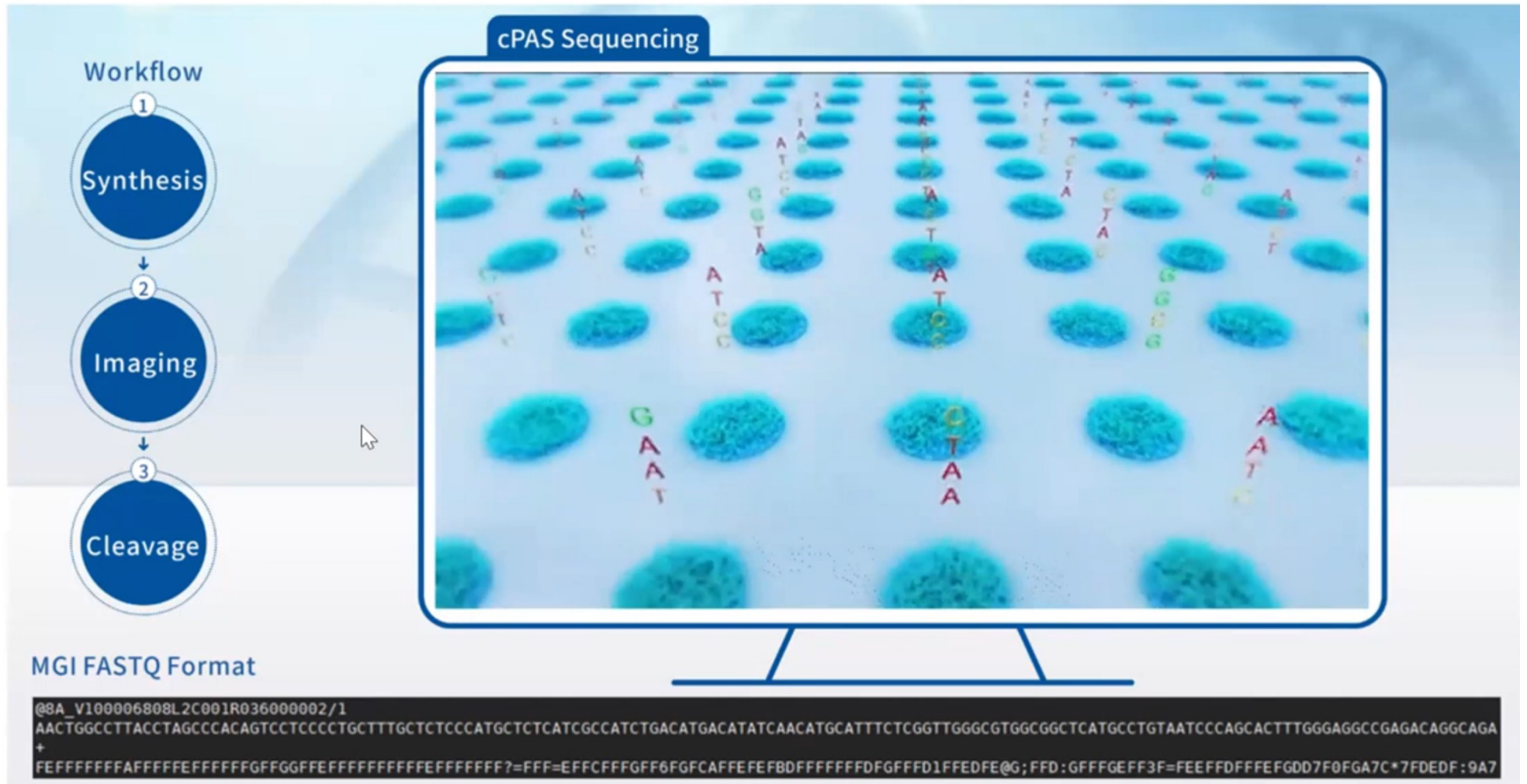
One ssCirDNA molecule produces a weak signal during sequencing



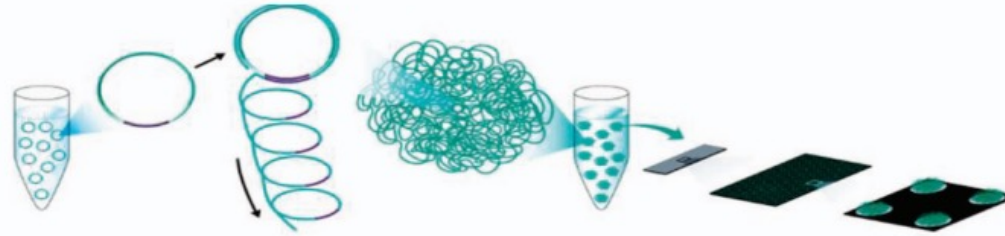
A DNB (300-500 copies of the ssCirDNA) produces a stronger signal for base calling



Primary Analysis: Base-calling



A: DNBseq™



B: CoolNGS™

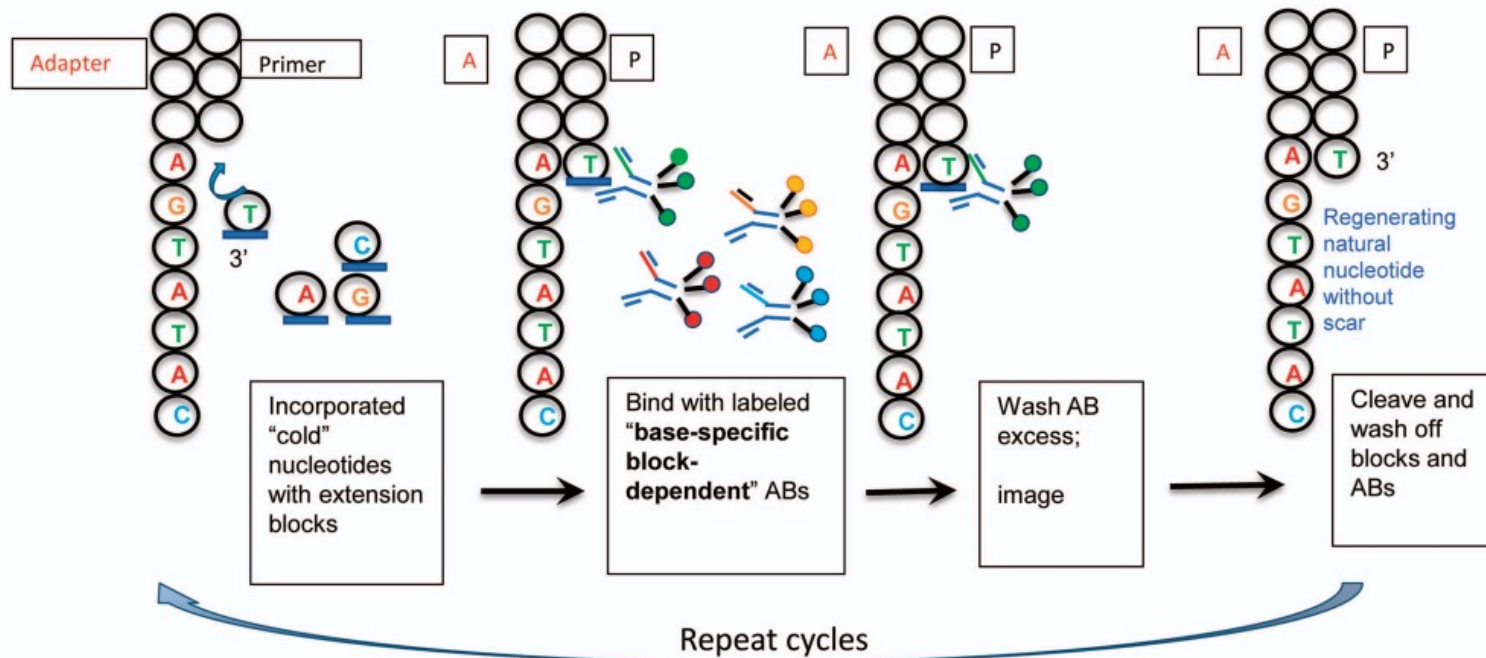


Fig. 3. DNA nanoballs and combinatorial CoolMPS.

(A), DNA nanoballs: the generation of DNA nanoballs by ligation of individual DNA molecules to an adenoviral vector. This is followed by rolling circle replication of that template using a $\phi 29$ polymerase. The linear molecules then collapse into a nanoball that can be deposited into a patterned flow cell for sequence interrogation. (B), CoolMPS [combinatorial probe-anchor synthesis (cPAS)]: In a sequencing cycle, bases are determined by utilizing cPAS. First cPAS incorporates a cold nucleotide that has an extension block. The sequence of the actual nucleotide is determined by binding with labeled block-dependent antibodies. After each cycle, 2 wavelengths of laser (green and red) are collected by cameras. After imaging the signals, the antibodies and the blocks are cleaved and the process is then repeated successively to determine the sequence at one end of each

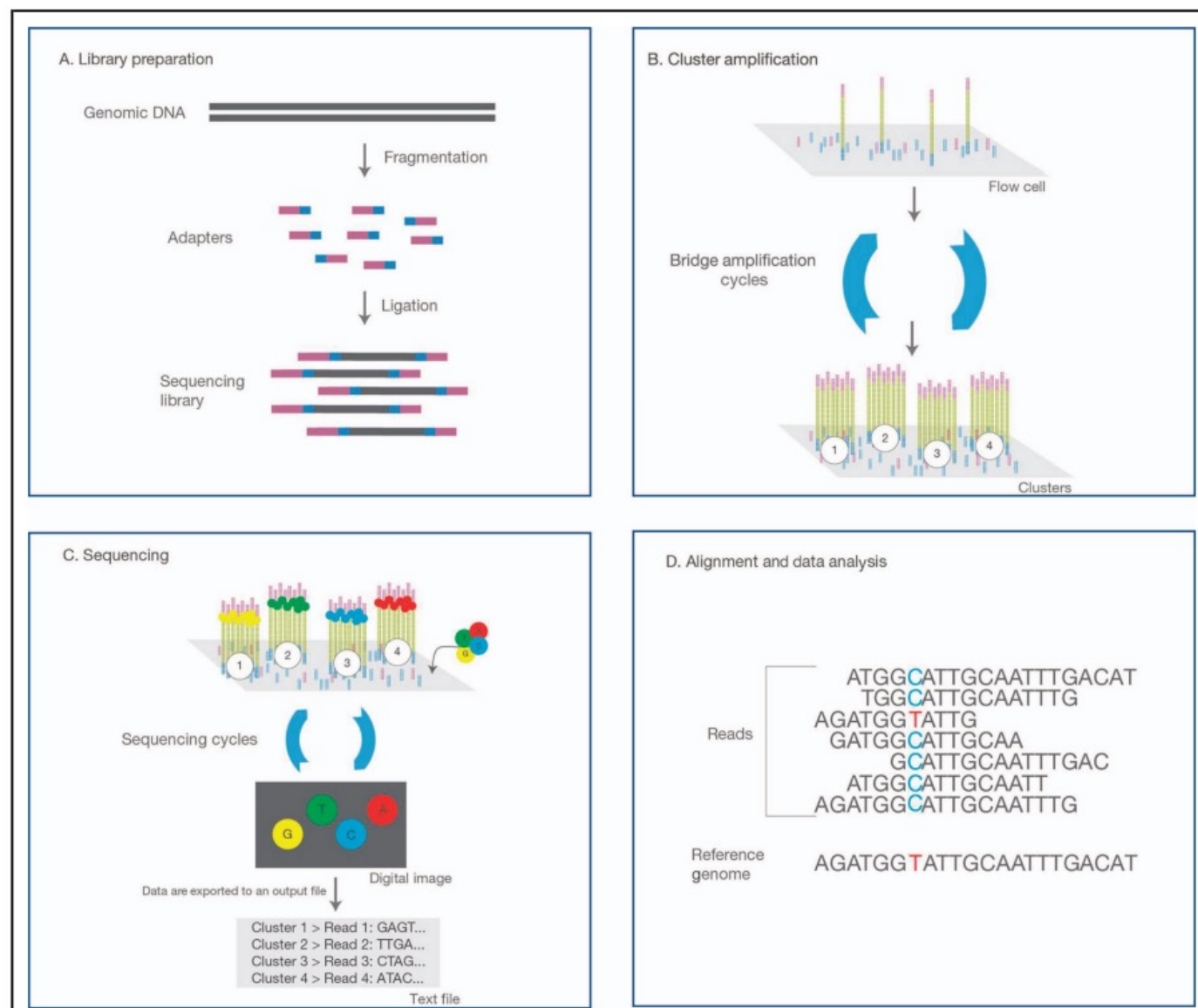
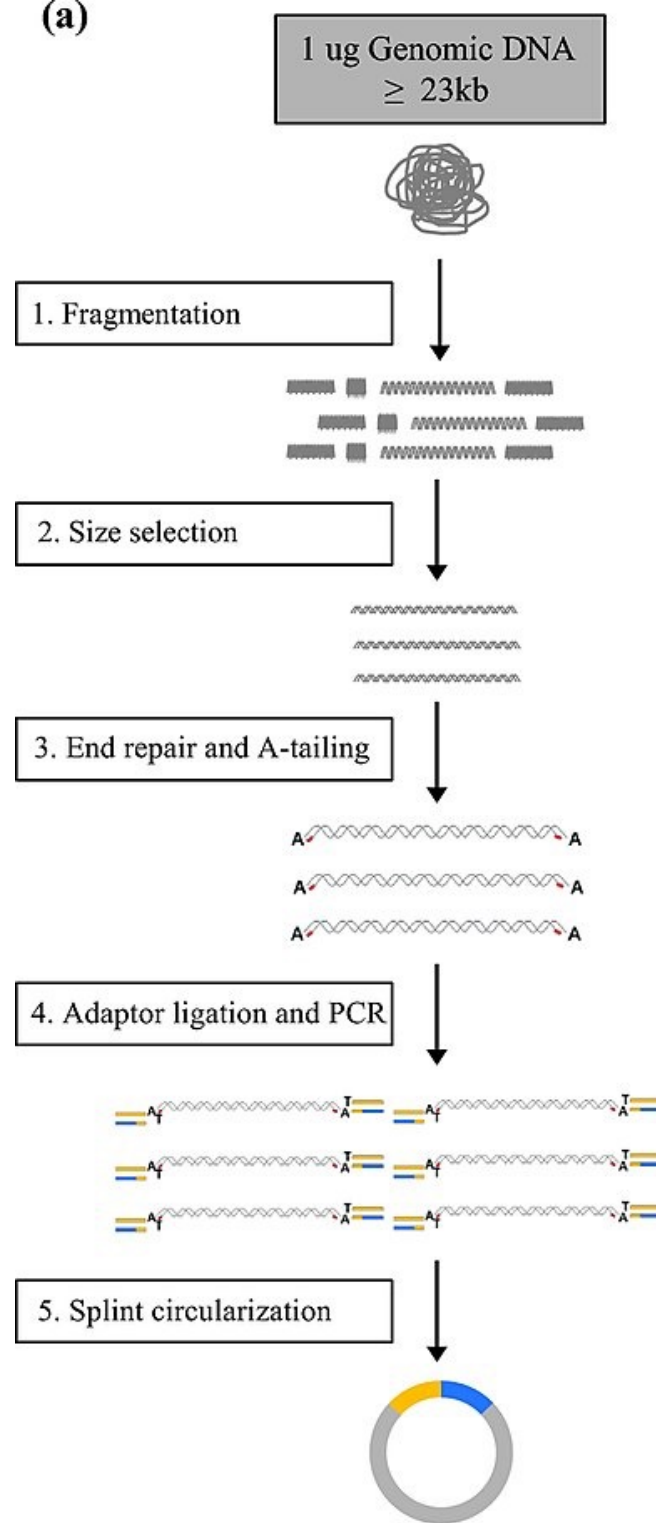


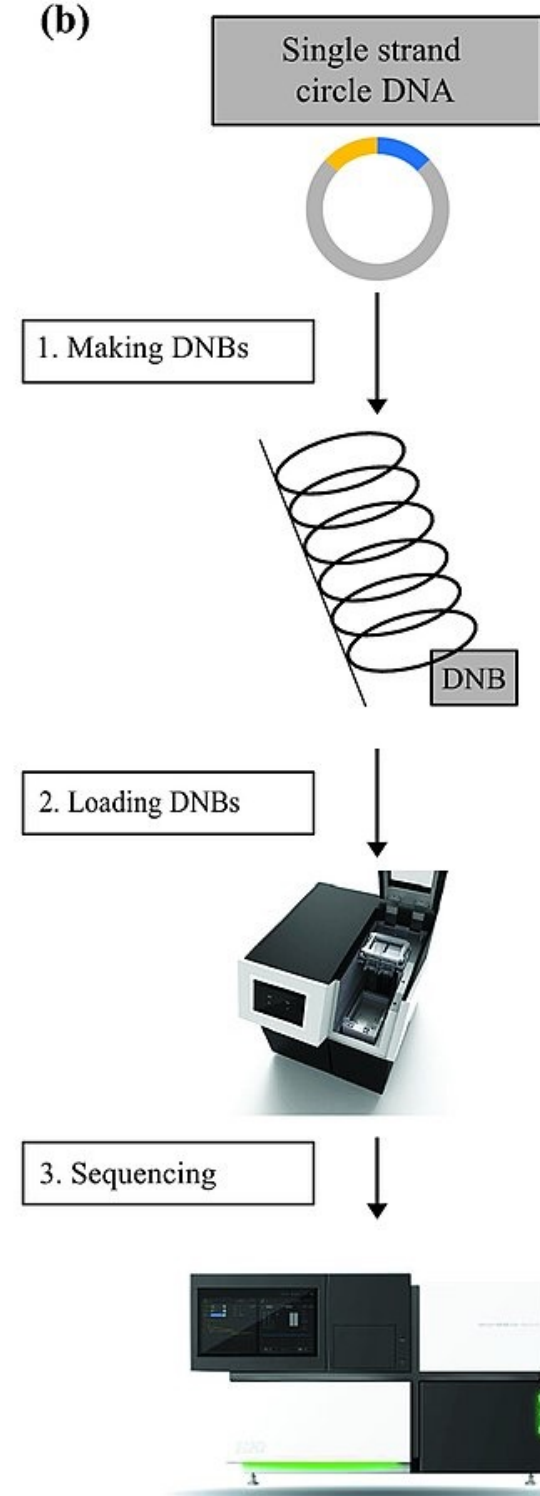
Fig. 2. Illumina sequencing platform.

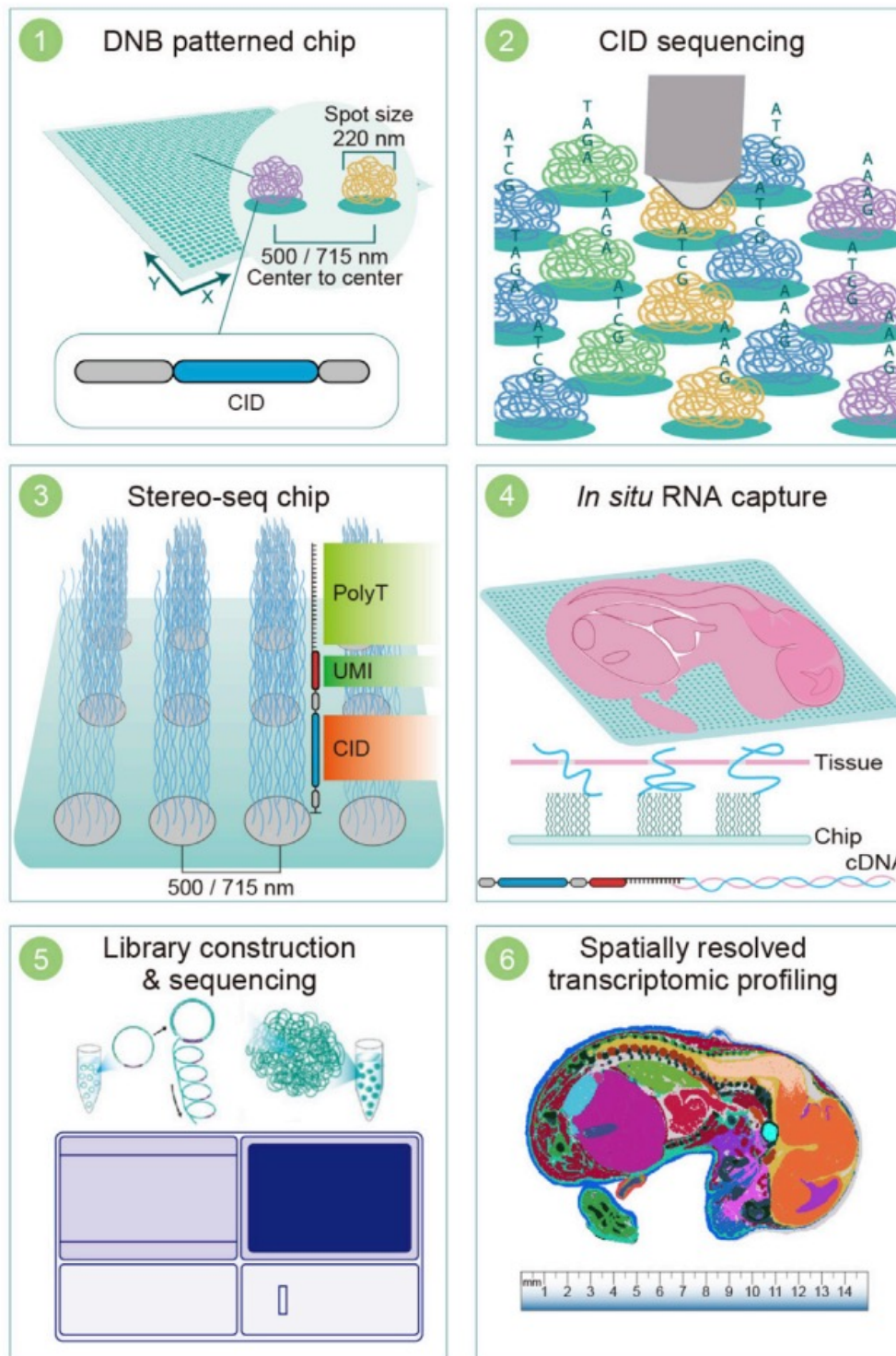
(A), NGS libraries are prepared by the fragmentation of genomic DNA and the ligation of the specified adaptors to both ends. (B), Library is loaded into a flow cell, and the fragmented DNA hybridize to the flow cell surface. Each bound fragment is amplified into a clonal cluster through bridge amplification. (C), Sequencing reagents, including fluorescently labeled nucleotides, are added to the flow cell, and the first cycle is imaged, and the emission from each cluster is recorded. The emission wavelength and intensity are used to determine the base called "n" times to create a read length of "n" base. (D), Reads are aligned to a reference sequence with a software tool. Differences between the reference genome and the newly sequenced reads can be identified. Image

(a)



(b)





Cell

Resource

Spatiotemporal transcriptomic atlas of mouse organogenesis using DNA nanoball-patterned arrays

Graphical abstract

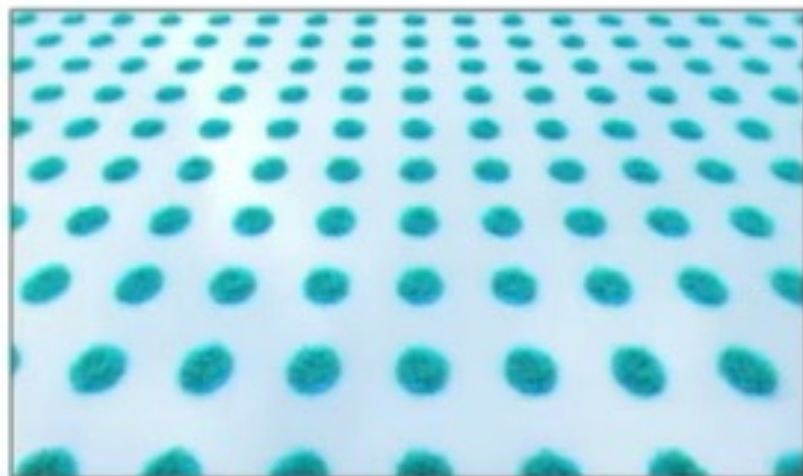
Stereo-seq
DNB patterned chip

Spatial transcriptomics at cellular resolution

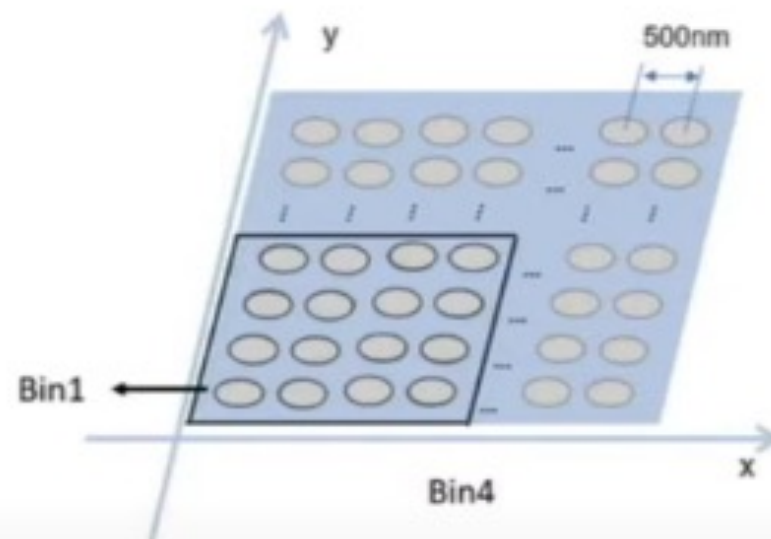
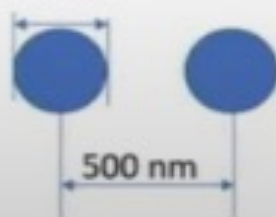
Authors

Ao Chen, Sha Liao, Mengnan Cheng, ...,
Longqi Liu, Xun Xu, Jian Wang

the coordinate identity (CID)



220 nm



Bin20: diameter = 10 μm

Bin50: diameter = 25 μm

Bin200: diameter = 100 μm